Overvalued Geographic Familiarity in International Asset Management

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Abstract

International equity funds increasingly hire foreign-educated managers. Within funds over time and across similar funds, greater foreign-educated representation is associated with lower factor-adjusted net returns. Their stock selection is weaker in markets geographically close or socially connected to their education countries, where their familiarity is expected to confer an advantage. Selected high-conviction positions in education countries earn high alphas, but they make up a small share of the portfolio and add little to its overall alpha. We connect these investment outcomes to how funds select and evaluate managers. Foreign-educated recruits arrive with weaker prior team records than U.S.-educated recruits. After appointment, their retention improves relative to that of U.S.-educated colleagues as fund performance worsens. Together, the findings suggest that funds overvalue geographic familiarity when assessing investment ability.

Keywords: Manager selection; local expertise; familiarity; manager education; international mutual funds; stock selection.
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1 Introduction

Information is local, and the frictions it creates are largest in international investment. Even within the United States, fund managers earn higher returns on nearby companies (Coval and Moskowitz, 2001). Across countries, investors hold far less foreign equity than diversification would suggest and are typically less informed than local investors about distant firms, institutions and markets (French and Poterba, 1991; Kang and Stulz, 1997; Brennan and Cao, 1997; Van Nieuwerburgh and Veldkamp, 2009). A natural remedy is to delegate foreign investment to managers who know those markets. International equity funds, a main vehicle through which U.S. investors hold foreign stocks, entrust investments across many markets to small teams of managers, and increasingly to managers educated abroad, whose knowledge of local languages, institutions and companies can provide a home-field advantage (Jagannathan et al., 2022). But knowing one market need not imply superior investment ability across an international portfolio, and because noisy returns and shared team records make individual ability hard to observe (Fama and French, 2010; Chevalier and Ellison, 1999a), funds must rely partly on visible qualifications such as where a manager studied. Does managers' geographic familiarity overcome international information frictions and add value, or do funds overvalue it when selecting and evaluating managers?

We find that funds with more foreign-educated managers earn lower factor-adjusted net returns. Outside managers' education countries, geographic, language and social ties are associated with less favorable stock selection for foreign-educated teams than for U.S.-educated teams. Foreign-educated teams themselves pick stocks worse in bordering and more socially connected markets, where familiarity should help most. Funds also overweight managers' education countries, but foreign-educated teams show no clear average stock-picking advantage there. Selected education-country positions nevertheless earn high alphas, indicating useful but narrowly concentrated local expertise. These positions account for less than 4% of the portfolio and add little to portfolio alpha at the country weights funds actually choose. Because this investment value is so narrow, we ask how funds value the background when

they hire and retain managers. Foreign-educated recruits bring weaker records from their previous fund teams than U.S.-educated recruits, even in the countries where their schooling should provide specialized knowledge. After appointment, their retention improves relative to that of U.S.-educated colleagues as fund performance worsens. Together, the findings suggest that funds overvalue geographic familiarity when hiring and retaining managers, treating it as a broader signal of investment ability than the performance evidence warrants.

We study 1,663 U.S. actively managed international equity funds over 2000–2024, linking managers' education and dated fund assignments to portfolio holdings and returns. Foreign-educated managers have at least one recorded degree outside the United States; the U.S.-educated comparison group has recorded degrees only in the United States. The funds operate within a common U.S. asset-management market but invest across many foreign countries, with a median of 19 per fund. This setting lets us assess the geographic scope of investment performance through country allocation, stock selection, and overall fund returns, alongside recruits' prior team records and the sensitivity of subsequent departures and new assignments to fund performance.

We begin with overall fund performance. We compare each fund with itself as its team changes over time, and with funds that share its benchmark in the same month. After adjusting for market and style exposures, a change from an entirely U.S.-educated to an entirely foreign-educated team is associated with net returns about 70 basis points a year lower. The gap is nearly identical in gross returns, so fees do not explain the shortfall.

We next examine stock selection outside the countries where current managers studied. To measure stock picking, we compare the returns funds earn on their investments in the same country and quarter, adjusted for stock characteristics. We test whether familiarity with a market improves stock selection. If studying in a country also conveys knowledge of nearby or closely connected markets, foreign-educated teams should pick stocks better in markets close to their education countries, where their background would be expected to add the most value. We measure closeness in three ways: a shared border; a common official

language; and social connectedness based on Facebook friendship links (Bailey et al., 2018). We find the opposite. Foreign-educated teams select stocks 184 basis points a year worse in markets bordering an education country than in other outside markets, and their selection declines as social connectedness to an education country rises. U.S.-educated teams show no such disadvantage in markets close to the United States. By all three measures, closeness is significantly less favorable for foreign-educated teams than for U.S.-educated teams; for a shared border, the difference is 484 basis points a year. Familiarity thus accompanies worse, not better, stock selection, consistent with familiarity bias (Huberman, 2001; Pool et al., 2012).

We then turn to the countries where managers studied. Relative to their benchmarks, funds put about 1.1 percentage points more weight in a country when one of their current managers studied there, and this overweight rises when the team gains a manager educated there and falls when it loses the last one. Managers' educational geography thus shapes where funds invest. Using the same stock-picking measure as above, however, foreign-educated teams show no clear average advantage over U.S.-educated teams within their education countries: the estimated difference is 0.2 basis points a year and statistically insignificant. The local advantage is instead concentrated. The positions that funds overweight most heavily within an education country earn a 4.0% annual alpha, 3.6 percentage points more than other positions in the same countries and 3.9 points more than similarly overweighted positions outside education countries, with every portfolio reweighted to the same country mix. At the country weights funds actually choose, however, the alpha of these positions falls close to zero, and they make up only 3.6% of funds' observed equity holdings, so their contribution to portfolio alpha is small and statistically insignificant. The local expertise is genuine but narrow: it appears in stock choices within education countries, not in overall fund performance.

If funds overvalue geographic familiarity, the overvaluation should appear in personnel decisions as well as in returns. Funds increasingly hire foreign-educated managers: their

share of new appointments rose from 32.3% in 2000–2004 to 42.5% in 2020–2024. We ask whether this demand is supported by demonstrated local skill, fundraising value or a broader assessment of the qualification. Stronger prior performance in the relevant country could compensate for a weaker general record; stronger fundraising could offer a commercial reason to accept it. We therefore test whether weaker general records are accompanied by better performance in the relevant country or by stronger fundraising, and whether observed local performance explains subsequent career responses.

We measure a recruit's record by the stock selection of the fund teams the recruit served on during the four quarters before the appointment. Including both education and other non-U.S. countries, the prior teams of foreign-educated recruits selected stocks 86 basis points a year less successfully than those of U.S.-educated recruits hired by the same fund, controlling for previous-fund benchmarks, school rankings, and other observed qualifications. Outside-country checks also yield negative gaps when performance is measured relative to funds sharing the same benchmark, country, and quarter, and when comparing recruits hired in the same round. Nor do they bring stronger fundraising records: their previous funds attracted weaker flows, and funds employing more foreign-educated managers show no clear gain in subsequent flows or fees. Thus, foreign-educated recruits arrive with weaker investment records and without a stronger fundraising record.

The specialist explanation predicts stronger prior performance in the market where a recruit's knowledge is needed. We test this prediction in countries the hiring fund already holds but where none of its continuing managers studied. In the countries needing coverage, the prior teams of recruits educated there selected stocks 274 basis points a year worse than those of U.S.-educated recruits hired by the same fund, measured in the same country and conditional on prior overall fund performance. Stronger demonstrated specialist records therefore do not explain the weaker records accepted at entry.

Finally, we examine how careers respond to performance after appointment. If funds give the qualification more weight and realized performance less, foreign-educated managers

should receive relative protection after poor returns and smaller relative rewards after good returns. We find these paired responses among colleagues on the same fund. When a fund's quarterly return falls ten percentage points relative to funds with the same benchmark, the probability that its foreign-educated managers leave falls by 1.45 percentage points relative to that of their U.S.-educated colleagues. When the return rises ten points, their probability of receiving a new fund assignment falls by 1.07 points relative to those colleagues. Both patterns persist after controlling for the performance of the managers' education-country holdings, so observed local performance does not account for their relative career responses.

Taken together, the evidence suggests that funds overvalue geographic familiarity as a signal of investment ability. The local knowledge it brings is genuine but concentrated in a few positions, while its broader use comes with worse stock selection in related markets, weaker records at hiring and relative career responses consistent with less weight on realized performance. This pattern resembles the "kernel of truth" in stereotypes described by Bordalo et al. (2016): a characteristic with a real but narrow advantage is treated as evidence of broad ability.

The paper contributes to research on the scope of investment information. Geographic, educational and offshore connections can improve investment decisions (Coval and Moskowitz, 2001; Cohen et al., 2008; Baik et al., 2010; Bai et al., 2022), while familiarity and perceived competence can affect portfolios without a corresponding advantage (Heath and Tversky, 1991; Huberman, 2001; Grinblatt and Keloharju, 2001; Graham et al., 2009; Pool et al., 2012). Relatedness also shapes international allocations (Schumacher, 2018; Kuchler et al., 2022). We connect these questions by comparing broad fund performance and choices in related markets with the concentrated gains in local high-conviction positions. The concentration of value complements evidence on industry specialization, active share and best ideas (Kacperczyk et al., 2005; Cremers and Petajisto, 2009; Antón et al., 2021).

The personnel evidence contributes to research on manager selection and evaluation. Education and background predict fund outcomes (Chevalier and Ellison, 1999a; Gottesman

and Morey, 2006; Chuprinin and Sosyura, 2018), and performance matters for replacement and reassignment (Khorana, 1996; Chevalier and Ellison, 1999b; Fang et al., 2014). Qualifications provide signals of productivity (Spence, 1973; Phelps, 1972; Aigner and Cain, 1977); subsequent outcomes allow employers to update those assessments (Farber and Gibbons, 1996; Altonji and Pierret, 2001), although fund returns are noisy (Fama and French, 2010). We bring these margins together by studying accepted general and country-specific records, later career responses and the geographic scope of investment performance. The result complements work on professionals' beliefs (Linnainmaa et al., 2021) by examining personnel and portfolio outcomes surrounding a particular qualification.

Our closest comparison is Jagannathan et al. (2022), who document a home-field investment advantage in international mutual funds. We ask whether funds credit that background beyond the markets where it helps, by connecting performance across the whole portfolio to how funds recruit and evaluate the managers who have it.

Section 2 develops the empirical questions, and Section 3 describes the data, summary statistics and the prevalence of foreign-educated managers. Section 4 proceeds from fund NAV to familiarity in outside countries, education-country allocation and education-country returns. Section 5 examines accepted recruitment records, demonstrated country expertise and later career responses. Section 6 concludes.

2 Institutional Setting and Empirical Framework

2.1 Institutional setting

International equity funds give U.S. investors access to equity markets abroad. Morningstar classifies a fund in this group when foreign holdings account for at least 40% of its equity portfolio on average over the preceding three years. In a typical sample quarter, about 760 active funds operate, with mean assets of $1.6 billion. Most have global or global-ex-U.S. benchmarks; others follow emerging-market, regional or single-country mandates.

The median fund holds roughly 80 stocks in 19 countries, compared with more than 5,000 stocks in about 50 countries in the Morningstar Global ex-US index. This setting lets us distinguish the allocation across countries from the selection of stocks within each country, while observing the overall return investors receive.

Investment decisions are delegated to small teams. The average fund has 3.1 managers with mean tenure of 5.6 years, and managers often serve on several funds simultaneously. Returns and holdings are observed for the team even when research responsibilities are divided among its members. We therefore use team education in the performance tests and the records of a recruit's previous teams in the appointment tests. Individual spells identify later departures and assignments. We use education countries to locate each manager's potential specialty.

The adviser or sub-adviser selects the managers assigned to a fund. Appointees may come from another firm, from the same family or from outside the recorded fund-management population. Experienced recruits bring observable fund-team records alongside the educational qualifications in their biographies. These records allow us to compare the performance accompanying appointments; subsequent spells show how colleagues' careers vary with shared fund performance. The two comparisons capture entry and later evaluation within the same organizational setting.

2.2 Competing explanations and empirical implications

Foreign education can provide knowledge of institutions, business practices and local companies, or make further information gathering less costly (Coval and Moskowitz, 2001; Jagannathan et al., 2022). The question is how far that advantage extends. A country specialist can be useful without possessing superior investment ability across a fund's international mandate.

The overvaluation hypothesis concerns the extension of familiarity beyond its productive scope. Managers may treat connections to nearby or socially related markets as useful

information even when those connections accompany weaker stock selection. Fund families may treat a country specialty as evidence of broader investment ability when selecting and evaluating managers. We refer to this extension as overgeneralization, following Bordalo et al. (2016). Rational specialization offers a competing explanation for weaker general records: the family may value a sufficiently useful country specialty or a commercial benefit. The empirical tests therefore examine broad performance, recruits' demonstrated specialist and fundraising records, and the role of local performance in career evaluation.

The first set of tests concerns investment performance. If foreign education signals broad investment ability, funds with more foreign-educated managers should earn higher factor-adjusted NAV returns, which include any gains in education countries. If its value is local, any advantage should appear in education-country stocks rather than across the portfolio. If familiarity is overgeneralized, foreign-educated teams should select stocks no better, and possibly worse, in markets related to their education countries, where familiarity would seem most useful. We test these predictions with NAV, with stock selection outside and within education countries, and with high-conviction portfolios whose actual capital weights show how much local gains add to the portfolio.

The second set concerns the records funds accept at recruitment. If familiarity substitutes for demonstrated performance in recruitment, foreign-educated appointees can enter with weaker general records. Two alternatives could justify that trade-off: a stronger demonstrated country specialty, or a stronger fundraising record. We therefore compare accepted recruits' general investment and fundraising records, and then their prior selection in the countries where the hiring fund lacks an incumbent schooling link. The first comparison establishes the accepted-record difference and addresses the commercial alternative; the second asks whether superior observed specialist performance explains the difference.

The third set examines how funds respond to performance after appointment. Suppose a family's assessment combines a qualification-based view of a manager with the evidence in realized returns. Placing more weight on the former and less on common fund returns

predicts weaker career responses in both directions: greater relative retention as returns worsen and smaller relative gains in new assignments as returns improve. Common fund returns may also rationally receive less weight if they reveal less about a specialist's work. We therefore compare colleagues exposed to the same fund return and add performance in the observed local specialty. If local performance accounts for the different responses, conditioning on it should attenuate them. This connects the record accepted at entry to the performance used in later evaluation.

3 Data and Measures

3.1 Funds, holdings and returns

Our sample consists of U.S. actively managed international equity funds in Morningstar over 2000–2024, including funds that later liquidated or merged.[1] For each fund we observe quarterly stock holdings, the holdings of its assigned benchmark, monthly net returns and assets in U.S. dollars, reported gross returns, and annual expenses and turnover. Security returns and market capitalizations come from Datastream. We assign each company to the country of its primary business activity rather than its listing, which aligns country weights with economic exposure (Demirci et al., 2022).[2]

3.2 Manager education

Morningstar biographies list each manager's post-secondary institutions, and dated fund spells identify the managers serving together at each quarter-end, as well as appointments and departures. We map institutions to countries by hand and exclude professional cer-

[1] The extraction sets U.S. Category Group to International Equity, Oldest Share Class to Yes and Index Fund to No, without restricting the sample to surviving funds. Each fund is identified by the ISIN of its oldest share class, whose returns we use; identifiers that share a Morningstar fund identifier form one economic fund, the unit of clustering.

[2] Where total-return series are unavailable, returns are constructed from prices and dividends. Missing returns are excluded rather than set to zero.

tificates and executive programs.[3] A manager is foreign-educated if at least one recorded degree is from outside the United States and U.S.-educated if all recorded degrees are from the United States. Education is known for 85.5% of manager-quarters in the NAV sample.

Our main variable, ForeignShare_{jq}, is the fraction of fund j's managers at quarter-end q who are foreign-educated, among those with known education. The fraction with unknown education, UnknownShare_{jq}, enters separately.[4] Coefficients compare an entirely foreign-educated team with an entirely U.S.-educated team; a ten-percentage-point change is one tenth as large. The measure captures where managers studied, not their nationality or the diversity of the team (cf. Lu et al., 2024). Appointment and career tests use the individual indicator $\mathbb{1}_i^{\text{Foreign}}$, which equals one if manager i is foreign-educated.

For a given team, education countries are the non-U.S. countries in which at least one current manager studied, and we refer to holdings there as local. All other foreign markets are outside countries. In the country panel, $\mathbb{1}_{jcq}^{\text{Link}} = 1$ if a current manager studied in country c. Because a team can include foreign-educated managers without a link to a particular country, $\mathbb{1}^{\text{Link}}$ and ForeignShare capture different things. We measure how closely a market is related to an education country by shared borders and official language from CEPII GeoDist (Mayer and Zignago, 2011) and by the Facebook Social Connectedness Index (Bailey et al., 2018). For U.S.-educated managers, these ties are measured relative to the United States.

Except where noted, the regressions control for five observed team traits, covering gender composition, CFA charters, biography coverage and fund-management experience, to account for differences in team composition and experience.[5]

[3] A manager with degrees from several countries is linked to each. Conflicting profiles are coded as unknown education. Education is reconstructed from collected profiles; it is not an archive of the biography visible to an employer at the appointment date.

[4] For a team with two foreign-educated, one U.S.-educated and one unknown-education manager, the foreign-educated share is 2/3 and the unknown-education share is 1/4.

[5] The traits are the female share among managers with known gender, the CFA share among managers with biographies, the fractions with unknown gender and no biography, and mean log one plus years since the manager's first recorded fund spell. Appointment tests also include indicators for degrees from top-100 and 101–500 institutions in the 2003 Academic Ranking of World Universities, so ranking-controlled appointment comparisons begin in 2004.

3.3 Performance measures

Our first measure is the fund's net return, which captures the outcome for investors across the entire portfolio, including country allocation, stock selection, cash, trading and costs (Kacperczyk et al., 2008). To compare funds with different mandates and styles, we deduct the return explained by predetermined exposures to thirteen factors: the five factors and momentum for developed ex-U.S. markets and for emerging markets (Fama and French, 2015), plus the U.S. market. For fund j in month m,

$$r^a_{jm} = R^{\text{net}}_{jm} - R^f_m - \hat{\beta}'_{j,m}\mathbf{f}_m. \tag{1}$$

Here R^{net}_{jm} is the net return, R^f_m the one-month Treasury-bill return, $\mathbf{f}_m$ the factor returns and $\hat{\beta}_{j,m}$ the fund's loadings estimated over the preceding 60 months. Estimating loadings before the evaluated month lets exposures evolve and keeps the evaluated return out of its own loading estimate.[6] We express this factor-adjusted return in annualized basis points, $R^a_{jm} = 12 \times 10^4 r^a_{jm}$.

Our second measure evaluates stock selection within a country. A fund can benefit from a strong national market without picking better companies there; comparing its holdings with similar local stocks separates the choice of stocks from the choice of countries (Brinson and Fachler, 1985; Coval and Moskowitz, 2001). For each country c in which fund j holds stocks at quarter-end q, we compare the return on the fund's holdings in that country over the next three months with the return on a matched portfolio of similar local stocks, following characteristic-adjusted performance evaluation (Daniel et al., 1997; Wermers, 2000). Each stock k the fund holds receives a benchmark return. Within the country and quarter, stocks are sorted into terciles by market capitalization at the quarter-end and, within each size tercile, into terciles by their cumulative return over the eleven months before the quarter-

[6] Loadings require at least 36 observed returns. Without that history, we use loadings estimated on other funds with the same benchmark over the same window, excluding the focal fund, with broader peer groups where needed; 32,584 of the 206,847 main-sample months use peer loadings. The adjustment deducts only the factor component of the loading regression.

end month. The benchmark return $\tilde{r}_{km}$ in month m is the capitalization-weighted return of the other stocks in the same size and past-return group, excluding stock k itself.[7] In each of the three months m that follow quarter-end q, the return on the fund's holdings in country c in excess of these benchmarks is

$$r^{\text{adj}}_{jcm} = \frac{\sum_k \omega_{jkm} \left(r_{km} - \tilde{r}_{km}\right)}{\sum_k \omega_{jkm}}, \tag{2}$$

where r_{km} is stock k's return in month m, and ω_{jkm} is stock k's weight in the fund's portfolio at quarter-end q, grown with the stock's returns from the quarter-end through month $m-1$. The sums run over the fund's stocks in country c that have a return in month m; stocks with missing returns are dropped rather than set to zero. We refer to the fund's holdings in a country as its country basket. The fund's adjusted country return, R_{jcq}, is the average of r^{adj}_{jcm} over the three months, annualized and expressed in basis points per dollar of the basket. A positive value means that the fund's holdings in country c beat similar local stocks. Holdings and teams observed at quarter-end q are thus paired with returns over the following quarter, and both are updated each quarter. The same measure underlies the prior-record tests in Section 5.

NAV coefficients describe performance per dollar of fund capital; country-return coefficients describe performance per dollar invested in a particular market. The country tests locate geographic stock-selection patterns within the broader NAV comparison. Because the regressions use different comparisons and weights, their coefficients are not components of the NAV coefficient. Appendix Table A21 summarizes the outcomes, units and timing.

[7] The comparison stocks are all stocks held by sample funds in that country and quarter, and their capitalization weights drift with returns within the quarter. When fewer than five other stocks in the group have a return that month, the benchmark uses the size tercile alone and, failing that, the country's market return; about three quarters of holdings capital receives the full size and past-return match.

3.4 Sample and summary statistics

All analyses start from the same 1,663 funds with manager records, an assigned benchmark and valid country weights, and each test uses the observations its outcome requires (Appendix Table A1). The main NAV sample has 206,847 fund-months at 1,418 funds with at least 24 months of eligible return history; including all 1,536 funds with usable returns gives a similar coefficient. The joint country-selection panel has 974,924 fund-country-quarters at 1,428 funds, including 38,806 in education countries; the outside-country panel used in Section 4.2 has 936,076 at 1,423 funds.

Table 1 reports summary statistics. The average fund has $1.6 billion in assets, annual expenses of 1.22% and about three managers. The mean foreign-educated share is 0.35, and about half of fund-quarters have no foreign-educated manager. Among the 2,992 managers with recorded education, 26.9% studied only outside the United States, and the United Kingdom is the most common foreign education country. Figure 1 shows that the fraction of funds with at least one foreign-educated manager rose from 27.8% in 2000Q1 to 49.5% in 2024Q4.[8]

Panel standard errors are clustered by fund, allowing for dependence over time and across a fund's countries (Petersen, 2009; Cameron and Miller, 2015). Unless stated otherwise, fund-months and fund-quarters receive equal weight; precision-weighted estimates are supplementary checks (Solon et al., 2015). Portfolio alphas use monthly time series.[9]

4 Investment Performance and Geographic Familiarity

We organize the investment evidence around the scope of geographic familiarity. We first measure the fund's overall NAV performance, then examine stock selection in outside coun-

[8]Each fund receives equal weight, and funds with unknown education remain in the denominator; because the figure covers all active sample funds, its shares differ from the NAV-sample statistics in Table 1. Appendix A.1 examines the foreign-educated share of new appointments.

[9]Country-allocation regressions cluster by fund identifier. Portfolio regressions use a monthly risk-free rate converted from the Treasury-bill yield, conventional standard errors for individual alphas and robust Wald tests for differences between alphas.

tries with differing ties to managers' education countries. We next turn to the education countries themselves: how much capital funds allocate there, how their stock selection compares with U.S.-educated teams, and whether any gains are concentrated in high-conviction positions.

4.1 Fund returns and expenses

The NAV comparison asks whether greater foreign-educated representation accompanies better performance across the entire fund. The outcome R^a_{jm} is the factor-adjusted net return of Equation (1), in annualized basis points. We follow the same fund as its team changes and compare it with funds sharing its benchmark in the same month:

$$R^a_{jm} = \alpha_j + \lambda_{bm} + \rho \text{ForeignShare}_{j,q-1} + \Gamma' X_{j,q-1} + u_{jm}. \tag{3}$$

Here q is the quarter containing month m, so the team at the preceding quarter-end determines ForeignShare for the three months of quarter q and education precedes the evaluated returns. Fund effects α_j account for persistent differences across funds; benchmark-month effects λ_{bm}, where b is the fund's assigned benchmark, compare funds facing the same mandate and date. The coefficient ρ is the annualized return difference associated with a zero-to-one change in the foreign-educated team share.

Staffing can respond to the fund's recent performance and business conditions. We control for those conditions using prior-year benchmark-relative returns and volatility, preceding-quarter flows and lagged assets. The remaining controls capture team scale and observed qualifications, expenses, turnover and top-ten holdings concentration.[10] Because portfolio

[10] X contains unknown education; log team size and log lagged assets with spline terms; prior-year relative performance and volatility; preceding-quarter flows; prior-year expenses and turnover; prior-quarter top-ten concentration; and the five team traits defined in Section 3. Prior-year returns require nine observed months and use benchmark peers excluding every share class of the focal fund. Volatility is log one plus annualized relative volatility; flows enter directly in percentage points of assets at the beginning of the preceding quarter, with a missing-flow indicator. Expenses are annual basis points, turnover is log one plus the reported percentage divided by 100, and concentration is the observed top-ten equity weight. Missing-value indicators retain unavailable controls.

operations may themselves reflect earlier staffing decisions, we also report the model without expenses, turnover and concentration. Fund-months receive equal weight, with standard errors clustered by economic fund.

Funds earn lower factor-adjusted net returns when a larger share of their managers is foreign-educated. Table 2, column 1, estimates −69.5 annual basis points for the full zero-to-one difference in the team share, with a standard error of 35.4. At the sample mean share of 0.35, the fitted difference from an otherwise comparable zero-share observation is approximately 24 basis points a year. For scale, the full-share coefficient exceeds half the sample's average annual expense ratio of 1.22%. The sample and operating-control comparisons in Table 2 and Appendix Table A9 give estimates between −66 and −70 basis points. Weighting funds by the precision of their returns gives −70 with $p = 0.002$. These comparisons show a similar economic magnitude when supported observations or their relative weights change; the equally weighted model remains the primary estimate. Because NAV includes education-country investments, the estimate already incorporates any local gains.

We evaluate factor-adjusted net returns, accounting for systematic market and style exposures when assessing active management (Carhart, 1997; Fama and French, 2010). Appendix Table A8 reports the comparison with precision weights.

To ask whether the shortfall arises before or after expenses, we replace net NAV with reported gross NAV in Equation (3), keeping exposures, observations, controls, effects and weights fixed. On 205,773 common fund-months, both coefficients are approximately −71 annual basis points (Table 3). Their paired difference, $\rho_{\text{gross}} - \rho_{\text{net}}$, is −0.47 basis points, with a 95% interval of $[-2.45, 1.51]$. This interval is narrow relative to the return gap, and the comparison without operating controls gives the same conclusion: the shortfall arises in gross returns, not in the expenses deducted from them.

A possible explanation for weaker fund performance is that foreign-educated managers' country knowledge is poorly matched to their mandates. If matching is important, their performance should be more favorable when their education countries account for more of

the benchmark. We measure this relevance using all recorded degree levels, as in the main analysis. For each appointment, we sum benchmark weights in the manager's distinct non-U.S. education countries, average them over the four quarter-ends before appointment, and hold the measure fixed throughout the spell. We then sum these relevance measures across current foreign-educated managers and divide by the number of managers with known education. This team measure increases with the benchmark importance of managers' education countries, rather than with the fund's subsequent choice to overweight them.[11]

We augment Equation (3) with this pre-appointment relevance measure and the foreign share with unknown relevance, retaining total ForeignShare, the original controls, fixed effects, weights and 206,847 observations at 1,418 funds. The relevance coefficient, θ, therefore asks whether more relevant schooling predicts better returns at a given foreign-educated share. Appendix Table A7 reports both the original controls and a specification that also holds the team's foreign education-country composition constant. The latter separates relevance from differences in where managers studied. The relevance coefficient is -463.2 (standard error 295.1) with the original controls and -688.3 (standard error 244.6), significant at the 1% level, with country-composition controls. Among appointments with measured relevance, moving from the lower to upper quartile raises benchmark exposure to education countries from 0.6% to 11.1%. For a fully foreign-educated team with measured relevance, the second estimate implies 72.3 basis points lower annual factor-adjusted NAV returns; at a fixed foreign share of one-half, it implies 36.1 basis points.

This is a relevance gradient conditional on foreign representation, not another estimate of replacing U.S.-educated managers or an additional loss to add to the main NAV coefficient. Greater measured relevance at appointment is associated with worse, rather than better, subsequent performance after accounting for education-country composition. The result challenges the simple explanation that geographic knowledge fails only because it is deployed

[11] We use historical country weights of the fund's recorded benchmark; its assignment is not a dated mandate history. A separate share identifies foreign-educated managers whose relevance cannot be measured over all four pre-appointment quarters. They remain in the estimation sample. Appendix B.1 gives the construction and specification.

in unsuitable mandates. It strengthens the case for examining whether funds overvalue geographic familiarity when selecting and evaluating managers in Section 5; it does not by itself identify their beliefs or separate those beliefs from hiring in response to difficult investment conditions.

Additional specifications examine portfolio controls, staffing timing and performance beyond the initial appointment period (Appendices B.3, B.4 and B.5). Team composition four quarters ahead adds little: its coefficient is −3 basis points (standard error 37), and the current-share coefficient stays near −72. The shortfall is also present for managers beyond their first two years at the fund (−81 basis points, standard error 38). The holdings tests next examine whether the fund-return shortfall is accompanied by unfavorable stock selection where familiarity should help. They compare investments within the same country and quarter, starting outside managers' education countries.

4.2 Stock selection and familiarity outside education countries

We first examine countries where no current manager studied.[12] If studying in a country also conveys knowledge of nearby or closely connected markets, foreign-educated teams should select stocks better in markets close to their education countries in geography, language or social ties, where familiarity should help most. If familiarity instead brings confidence without useful information, their selection in those markets should be no better, or worse. We compare how relatedness is associated with stock selection for foreign- and U.S.-educated teams.

We measure how closely a market is tied to where a manager studied in three ways, each in a separate regression: a shared border; a common official language; and the log of Facebook social connectedness from the October 2021 snapshot. For foreign-educated managers, the tie is measured between the market and the manager's non-U.S. education countries, using the closest tie when there are several. For U.S.-educated managers, it is

[12] Appendix B.6 shows how the foreign-versus-U.S. comparison in these countries changes over the sample period.

measured between the market and the United States. Both groups therefore face the same question: is stock selection better in markets close to where the manager studied?

Ties are defined for managers, but returns are observed for the team, so we average the ties over the team. Let z_{ic} measure how closely market c is tied to where manager i studied: a zero–one indicator for border and language ties, or log social connectedness. For fund j at quarter-end q, with N_{jq} managers of known education, we form two team averages:

$$\text{Tie}_{jcq} = \frac{1}{N_{jq}} \sum_i z_{ic}, \qquad \text{ForeignTie}_{jcq} = \frac{1}{N_{jq}} \sum_i \mathbb{1}_i^{\text{Foreign}} z_{ic},$$

where both sums run over the team's managers with known education and $\mathbb{1}_i^{\text{Foreign}}$ equals one if manager i is foreign-educated. Tie averages every manager's tie; ForeignTie averages the same ties but counts U.S.-educated managers as zero. For example, consider a team with one manager educated in Germany and one educated only in the United States. For its holdings in Austria, which borders Germany but not the United States, Tie = ForeignTie = 1/2. For its holdings in Canada, which borders the United States but not Germany, Tie = 1/2 and ForeignTie = 0. ForeignTie is therefore an average of manager-level interactions, not the product of the team's foreign share and its average tie. For each tie measure, we regress the fund's adjusted country return in the market, R_{jcq}, on these team averages:

$$\begin{aligned} R_{jcq} = \alpha_{jc} + \lambda_{cq} + \gamma \text{ForeignShare}_{jq} + \delta \text{Tie}_{jcq} \\ + \theta \text{ForeignTie}_{jcq} + \Gamma' X_{jcq} + \varepsilon_{jcq}. \end{aligned} \tag{4}$$

Each coefficient has a direct interpretation. Raising a U.S.-educated manager's tie raises only Tie, so δ is the U.S.-educated slope: how selection in a market changes with its closeness to the United States. Raising a foreign-educated manager's tie raises both Tie and ForeignTie, so $\delta + \theta$ is the foreign-educated slope: how selection changes with closeness to the manager's education countries. The focal coefficient θ is the difference between the two slopes; it is negative if closeness is associated with less favorable selection for foreign-

educated managers than for U.S.-educated managers. Finally, γ is the foreign-versus-U.S. difference in markets with no tie to either group's education.[13] For the border and language measures, a unit change compares an entirely tied team with an entirely untied team; for social connectedness, slopes are per log point.

Fund-country effects α_{jc} compare the same fund in the same market over time, and country-quarter effects λ_{cq} remove returns common to all funds investing in that market in that quarter. Because each manager's tie to a market is fixed, the slopes are identified from changes in a team's composition within a fund and market, measured relative to other funds investing in the same market that quarter. The sample contains only markets outside the team's current education countries. Controls X contain $\text{MissingShare}_{jcq}$, log country stock count, log team size, unknown-education share and the five team traits.[14] Countries receive equal weight within each fund-quarter, and standard errors are clustered by fund.

Table 4, column 1, reports the border results. For foreign-educated managers, the slope $\delta + \theta$ is -184 basis points (standard error 67, significant at 1%): their teams select stocks worse in markets bordering an education country than in other outside markets.[15] For U.S.-educated managers, the slope δ is $+300$ basis points (standard error 135, significant at 5%): their teams select stocks better in markets bordering the United States. Only Canada and Mexico border the United States, whereas the foreign-education tie varies across many country pairs. The difference, $\theta = -484$ basis points (standard error 158), is significant at 1%: the same border tie is associated with selection 4.84 percentage points a year less favorable for an entirely foreign-educated team than for an entirely U.S.-educated team. These are returns per dollar of the country basket, not differences in whole-fund NAV.

[13] For social connectedness, γ refers to zero log connectedness, outside the observed range, so we interpret only the slopes in that model.

[14] Unavailable pair measures contribute zero to the tie averages, and $\text{MissingShare}_{jcq}$ records the share of managers whose tie cannot be classified: U.S.-educated managers with unclassified geographic ties in the border and language models, and foreign-educated managers with a missing social-connectedness pair in the social model.

[15] Standard errors of $\delta + \theta$ use the joint covariance of the two coefficients from the same regression.

The other two measures also yield a negative difference between the groups' slopes. With social connectedness (column 3), the foreign-educated slope is −43 basis points per log point (standard error 17, significant at 5%), and the difference from U.S.-educated managers is $\theta = -110$ basis points per log point (standard error 42, significant at 1%); for a one-standard-deviation increase in log connectedness, this difference amounts to 117 basis points a year. With a common official language (column 2), the difference is $\theta = -231$ basis points (standard error 67, significant at 1%). Most of it comes from U.S.-educated teams, whose selection is 155 basis points better in markets that share an official language with the United States (standard error 75, significant at 5%); the foreign-educated slope of −75 basis points is negative but not significant. Each measure enters a separate regression, so the estimates do not depend on conditioning one correlated tie measure on another.

These comparisons do not depend on the size and past-return benchmark. With raw country returns, which still compare funds holding the same country in the same quarter, the differences θ are −346 basis points for a shared border, −195 for a common official language and −81 per log point of social connectedness, significant at 5%, 1% and 5%, and the foreign-educated slopes remain negative (Appendix B.7). The familiarity pattern is therefore present in observed holdings returns as well as characteristic-adjusted returns.

Across the three comparisons, relatedness is less favorable for foreign-educated teams than for U.S.-educated teams, and foreign-educated teams themselves select stocks worse in bordering and more socially connected markets. This is the opposite of what useful familiarity would predict: selection is worse precisely where geographic familiarity should help most. We next turn from these related markets to the education countries themselves, examining first the capital allocated there and then the return on those investments.

4.3 Allocation to education countries

We next examine managers' education countries, beginning with whether funds put more capital in a country when one of their managers studied there. Because a large national

market receives a large weight mechanically, we measure each country's weight relative to the fund's assigned benchmark. For fund j, country c and month m, we estimate

$$\text{Overweight}_{jcm} = \alpha_j + \lambda_{cm} + \psi \mathbb{1}^{\text{Link}}_{jcm} + \Gamma' X_{jm} + e_{jcm}. \tag{5}$$

Here c ranges over non-U.S. countries, Overweight_{jcm} is the percentage of the fund's portfolio invested in country c minus that country's weight in the fund's assigned benchmark, in percentage points, and $\mathbb{1}^{\text{Link}}_{jcm}$ equals one if a current manager studied in c. The coefficient ψ measures how much more a fund overweights a country when one of its current managers studied there. A second specification replaces the indicator with LinkedShare_{jcm}, the fraction of the fund's current managers, including those with unknown education, who studied there, so that ψ compares an entirely linked team with an unlinked one. Country-month effects remove what all funds do in a country in a given month. We first include fund effects α_j and then replace them with fund-month effects α_{jm}; the latter compare countries within the same portfolio in the same month, so ψ then measures how a fund distributes its capital between linked and unlinked countries. Controls X contain the five team traits and are absorbed by fund-month effects.[16] The return comparisons that follow evaluate the investment value of this allocation.

Both measures show an active education-country tilt (Table 5). The indicator coefficients are 1.11 percentage points with fund effects and 1.09 with fund-month effects. The linked-share coefficients are 2.63 and 2.59, all significant at 1%. The latter compare a team with no linked manager with an entirely linked team; a ten-percentage-point increase in linked-manager share corresponds to about 0.26 percentage points more country overweight. The similar within-fund-month results locate the tilt in how funds distribute capital across

[16]The equally weighted panel starts with all 1,663 funds and contains approximately 5.6 million fund-country-months from January 1999 to December 2024; because these tests use only holdings and benchmark weights, not subsequent returns, they also include the 1999 holdings that precede the return sample; the fund-month models retain 1,661 funds after singleton fixed-effect groups are dropped. A missing-trait indicator retains teams without biographies, and the team record assigned to each month defines $\mathbb{1}^{\text{Link}}$ and X. Errors are clustered by fund identifier.

destinations. This establishes a preference; the return tests assess whether it accompanies an investment advantage.

Figure 2 traces the overweight around changes in the team. An event is the arrival of the first manager educated in country c, or the departure of the last such manager. With $\text{Overweight}_{jcq} = w_{jcq} - w^b_{jcq}$, where w_{jcq} and w^b_{jcq} are the fund's and its benchmark's percentage weights in country c at quarter-end q, we estimate separately for arrivals and departures

$$\text{Overweight}_{jcq} = \alpha_{jc} + \lambda_{cq} + \mu_{jq} + \sum_{\ell \neq -1} \psi_\ell \, \mathbb{1}\{q - q^*_{jc} = \ell\} + e_{jcq}. \tag{6}$$

Here q^*_{jc} is the quarter of the event for fund j and country c, so $\mathbb{1}\{q - q^*_{jc} = \ell\}$ equals one ℓ quarters after the event, for ℓ from -6 to 12, and each ψ_ℓ measures the change in the country's overweight relative to the quarter before the event, $\ell = -1$, which is omitted. Fund-country effects remove the fund's usual weight in that country, country-quarter effects remove changes common to all funds investing there, and fund-quarter effects remove changes in the fund's overall portfolio.[17] Weights rise about 0.5–0.7 percentage points after arrivals and fall 0.2–0.55 after departures, without a clear movement beforehand. Educational geography is therefore reflected in capital deployment as teams change. The next comparison asks whether stock selection in these education countries produces a return advantage over U.S.-educated teams.

[17]The 977 arrivals and 793 departures use up to twelve-quarter windows, a six-quarter clean pre-period, no overlapping events and truncation if the link reverses. The panels retain fund-identifier clustering.

4.4 Stock selection in education countries

The allocation evidence establishes an education-country preference, but does that preference accompany better stock selection? We use the same adjusted country return, R_{jcq}, as in the outside-country tests, now in a joint panel of education and outside countries:

$$R_{jcq} = \alpha_{jc} + \lambda_{cq} + \beta \mathbb{1}^{\text{Link}}_{jcq} + \gamma \text{ForeignShare}_{jq} + \Gamma' X_{jcq} + \varepsilon_{jcq}. \tag{7}$$

Here $\mathbb{1}^{\text{Link}}_{jcq}$ equals one if a current manager studied in country c, and ForeignShare_{jq} is the team's foreign-educated share. Two coefficients describe distinct comparisons. The coefficient γ is the difference between an entirely foreign-educated team and an entirely U.S.-educated team in markets where no current manager studied. The coefficient β is the additional difference when a current manager studied in the market. The comparison of interest, a fully foreign-educated team investing in its education country versus a U.S.-educated team investing in the same country, is therefore $\beta + \gamma$. As before, fund-country effects compare the same fund in the same market over time, and country-quarter effects compare funds investing in the same market and quarter, so β is identified when a link to a market begins or ends within a fund. Controls X contain log country stock count, log team size, unknown-education share and the five team traits. Countries have equal weight within a fund-quarter, and standard errors are clustered by fund.

The joint panel contains 974,924 fund-country-quarters at 1,428 funds, including 38,806 in education countries; links begin or end in 1,037 fund-country pairs at 525 funds.[18]

The joint model shows no clear average advantage in education countries. The comparison $\beta+\gamma$ is +0.2 annualized basket basis points, with a standard error of 56.6 (Table 6, column 1). Column 2 weights funds by the precision of their returns and gives a similar comparison (+5.6 basis points, standard error 51.4).

[18]The identifiers represent 1,429 oldest share classes. Formation quarters run from 1999Q4 through 2024Q3, with payoffs from January 2000 through December 2024. Country weights are renormalized over all supported foreign destinations within each fund-quarter. Unknown stock returns are excluded from covered baskets, not set to zero.

We next ask whether local information is concentrated in managers' largest active positions, and how those positions contribute under actual fund allocations.

4.5 The concentrated local value of geographic expertise

An average country basket can conceal information concentrated in a few companies. We therefore compare the positions managers overweight most with their ordinary holdings, first under a common country mix and then at actual capital weights. The first comparison evaluates selected stocks under benchmark country weights; the second measures their contribution to the pooled covered equity portfolio at actual weights.

4.5.1 High-conviction local positions

Country overweighting says where capital goes, but not which companies are well chosen. If local knowledge is most useful for particular companies, the advantage should be stronger in the fund's largest active stock bets. We identify those positions before observing their returns using a country-normalized version of the I-measure of Antón et al. (2021):

$$I_{jkq} = \frac{\omega_{jkq}}{\sum_{k' \in \Omega_{jcq}} \omega_{jk'q}} - \frac{\mathrm{MC}_{kq}}{\sum_{k' \in \Omega_{jcq}} \mathrm{MC}_{k'q}}. \tag{8}$$

Here k is a stock that fund j holds in country c at quarter-end q, Ω_{jcq} is the set of such stocks and k' runs over it, ω_{jkq} is the stock's weight in the fund's portfolio, as in Equation (2), and MC_{kq} is its market capitalization at quarter-end q. The measure subtracts the stock's capitalization share among those holdings from its weight in the fund's country basket. A positive value identifies an active overweight within the local stock selection. Normalizing within countries prevents a large country allocation from mechanically making every stock there a high-conviction position. High conviction here refers to revealed active weights among stocks the fund holds.

Within each fund-country-quarter, the top I-measure quartile defines high conviction and the other three quartiles define ordinary holdings. We form calendar-time portfolios for each group, separately in education and outside countries. Each portfolio pools the group's positions across all funds: within a country, positions are weighted by their dollar value, and countries are weighted by their shares in the Morningstar Global ex-US index. Reweighting to these benchmark country shares removes differences in aggregate national-market weights from the group comparisons. Each portfolio's alpha comes from the calendar-time regression

$$R_{pm} - R_m^f = \alpha_p + \beta_p' \mathbf{f}_m^{(10)} + \beta_p^G (R_m^G - R_m^f) + \varepsilon_{pm}. \tag{9}$$

Each regression uses the 300 months from January 2000 to December 2024. The portfolio index p identifies the country group and weighting rule; all local and outside conviction portfolios use this same model. All regression returns are monthly decimals: R_{pm} is the constructed portfolio's return, $\mathbf{f}_m^{(10)}$ contains the developed- and emerging-market five factors, and R_m^G is the return on the Morningstar Global ex-US index, with loading β_p^G. This common portfolio benchmark differs from the fund-specific assigned benchmarks in the NAV model. The focal intercept α_p is monthly excess performance beyond these exposures; multiplying it and its standard error by 12×100 gives annual percentage points, or by 12×10^4 gives annual basis points. Direct alpha differences use stacked portfolio series and robust Wald tests on intercept equality. The intercept evaluates the disclosed-holdings strategy.

The highest active local bets have a substantial alpha advantage. The top education-country quartile earns a 4.0% annual alpha (standard error 1.3), compared with 0.4% for ordinary local holdings (Table 7). Their direct 3.6-percentage-point difference is significant at 1%. Because both portfolios are reweighted to common benchmark country shares, this comparison evaluates stock selection within the same local investment opportunity set. It identifies a productive portfolio application associated with the education link.

Comparing top positions across country groups provides a second test of the local link. Top positions in education countries outperform top positions outside them by 3.9 percentage points a year ($p = 0.002$). Together with the 3.6-point top-versus-ordinary local difference, this contrast identifies a productive combination of an education link and an active stock bet. Figure 3 plots the four alphas, their confidence intervals and the direct tests. The top-versus-ordinary difference remains significant at 5% in six of the eight alternative constructions in Appendix Table A19, which vary the stock ranking, U.K. coverage, undergraduate links, risk models and trading costs; it is marginal with an expanded reference set ($p = 0.057$) and imprecise when benchmark weights define conviction ($p = 0.251$).

Figure 4 compounds the same monthly gross strategy returns over 2000–2024, starting with one dollar before January 2000. It shows when the local top portfolio separates from ordinary holdings and the benchmark, alongside the outside-country comparison. The curves track gross returns on the constructed strategies under benchmark country weights. The next test measures the payoff under the capital allocation funds actually choose.

4.5.2 The size of the local contribution

To assess economic scale, we measure these positions' contribution per dollar of pooled covered equity capital: the value of sample funds' disclosed stocks with usable returns, including U.S. stocks. Let s_m^L be the education-country share of covered equity capital in month m and R_m^L the return on those holdings at actual capital weights. Their monthly contribution to the covered equity return is

$$C_m^L = s_m^L R_m^L, \qquad R_m^{\text{all}} = C_m^L + C_m^O. \tag{10}$$

Here R_m^{all} is the return on the entire covered equity portfolio and C_m^O the contribution of the other covered equities. The annualized raw contribution is the average of C_m^L over the 300 months, times twelve, expressed in basis points. Weighting each month's return by that

month's capital share captures their comovement, which multiplying the average share by the average return would miss. To measure the contribution beyond risk exposures, we use the excess-return contribution $C^{\text{ex}}_{pm} = s_{pm}(R_{pm} - R^f_m)$ as the dependent variable in Equation (9), where s_{pm} is the share of covered equity capital in group p, all local, top local or ordinary local positions, and R_{pm} is now the group's return at actual capital weights. The intercept is then the alpha these positions add per dollar of pooled covered equity, rather than per dollar invested in them (Table 8). Education-country holdings represent 8.7% of covered capital and contribute 57.5 annual basis points of raw equity return. Their alpha contribution is −10 basis points (standard error 11): a positive realized equity payoff, but no clearly positive contribution beyond measured risk exposures.

High-conviction local positions occupy 3.6% of covered equity capital and have an actual-weight alpha contribution of 0.3 annual basis points (standard error 6.1). This differs from the 4.0% strategy alpha for two reasons. First, actual country weights change the return on the local stock portfolio: on a common sample of local holdings, the top local portfolio earns 406.7 basis points a year under the common country mix but 16.6 (standard error 159.3) at funds' actual country weights (Appendix Table A20). Second, the contribution series weights each month's return by its contemporaneous share of covered capital. The strategy alpha evaluates selected stocks under a common country mix; the contribution alpha measures their payoff at actual deployment.

The local evidence establishes a narrow benefit: high-conviction stocks perform well under a common country mix. At actual capital weights, the estimated contribution is small, distinguishing the value of selected local stocks from their payoff in the overall equity portfolio. Appendix C reconciles the weights and contributions.

The investment evidence distinguishes useful local expertise from a broad return advantage. Selected local stocks earn high alpha under a common country mix, yet funds with more foreign-educated managers earn lower factor-adjusted returns and select stocks worse in related outside markets. Meanwhile, foreign-educated appointments have become more

common (Appendix A.1). We next ask whether stronger fundraising (Section 5.1) or demonstrated country expertise (Section 5.2) explains the weaker general records accepted at entry, and whether subsequent fund returns have a different relative role in careers (Section 5.3). These tests examine whether funds credit geographic familiarity beyond its demonstrated payoff.

5 Recruitment and Career Evaluation

The investment results raise a personnel question: how do funds value geographic familiarity when hiring and evaluating managers? We first compare recruits' prior investment and fundraising records, then their records in countries needing expertise. We finally ask how colleagues' careers respond to the same fund performance. The hiring comparisons use accepted recruits' previous team records.

5.1 Records at appointment

Each observation is one manager's appointment to a fund. We measure the previous teams' average adjusted country return over the four full quarters before hiring, including both education countries and other non-U.S. countries. As in Section 3.3, these returns compare country holdings with similar local stocks. Including education countries allows any demonstrated local advantage to enter the recruit's overall international record; Section 5.2 separately examines the exact market needing expertise.[19]

Let h identify manager i's appointment to fund j in year y. The prior average return, $\bar{R}_h^{\text{pre}}$, is expressed in annualized basis points per dollar of country holdings; the bar denotes an average and pre means before appointment. An outside-country comparison, $\bar{R}_h^{\text{peer}}$, further subtracts returns of funds sharing the previous funds' benchmark, country and quarter (Ap-

[19] Countries have equal weight within origin fund-quarter. We average quarters, combine share classes of an economic origin and then average distinct origin funds. The characteristic-adjusted record covers non-U.S. stocks and is not a whole-fund NAV return.

pendix B.10). Thus, Y_h below is $\bar{R}_h^{\text{pre}}$ in the main regression or $\bar{R}_h^{\text{peer}}$ in the peer comparison. We estimate each separately:

$$Y_h = \alpha_j + \lambda_y + \delta \mathbb{1}_i^{\text{Foreign}} + \eta' \mathbf{B}_h + \Gamma' X_h + v_h. \tag{11}$$

The coefficient of interest is δ, on the indicator for a foreign-educated recruit. A negative δ means that foreign-educated recruits had weaker prior team records than U.S.-educated recruits accepted by the same fund, conditional on the controls. Hiring-fund effects α_j account for each fund's persistent recruitment standards, and appointment-year effects λ_y remove common recruitment conditions. The comparison uses hires across years within a fund. Because records partly reflect the previous funds' mandates, $\mathbf{B}_h$ controls for the benchmarks of the recruit's previous funds.[20] Controls X_h include school-ranking categories, gender, CFA, experience, the size and unknown-education share of the origin and destination teams, and the origin teams' country stock count. Appointments receive equal weight, and standard errors are clustered by hiring fund.

The comparison covers experienced appointees whose previous teams have observable records: 1,321 appointments at 300 hiring funds.[21]

Foreign-educated recruits arrive with weaker investment records. Table 9, column 1, estimates a difference of −86 annualized basket basis points (standard error 38), significant at 5%. Their previous teams' adjusted country returns are about 0.86 percentage points lower per year than those of U.S.-educated recruits accepted by the same hiring fund, accounting

[20] These are benchmark shares, weighted as in the record, for the 551 appointments with multiple origin benchmarks; Global ex-U.S. is the omitted category among 16. Prior NAV does not enter this equation, so the record is not conditioned on another realized performance measure.

[21] Of the appointments at the 1,663 funds, 1,600 have prior selection records, and support for the paired histories and the fund and year comparisons leaves 1,322 at 300 hiring funds. The appointment sample is held fixed at that of the earlier outside-country comparison, which required a same-benchmark peer and excluded one appointment. The current outcome restores all supported non-U.S. country baskets for these appointments, including education countries; it does not require peer-return availability. The same-round and needed-country tests use their own samples. Managers without a recorded prior fund role cannot enter this comparison.

for origin benchmarks and observed qualifications. The record includes education countries and predates the recruit's entry into the new fund.

An outside-country check further accounts for conditions facing the previous funds through $\bar{R}_h^{\text{peer}}$. On the same appointments and with the same regressors, the foreign-minus-U.S. gap is −72 annualized basket basis points (standard error 41), significant at 10% (Appendix Table A16). The difference therefore remains negative when the comparison uses other funds operating under the same benchmark, country and quarter conditions. The outside-country gap also remains negative among recruits hired by the same fund in the same quarter, which holds the fund's common appointment conditions fixed (−81 basis points, standard error 41; Appendix Table A17).

A weaker investment record could be acceptable if the recruit brings fundraising value. We assess this possibility first through prior fundraising records and then through subsequent fund flows. For the prior-record test in column 2, we use $Y_h = \text{Flow}_h^{\text{pre}}$ in Equation (11). This outcome is the quarterly net flow into the recruits' previous funds, in percentage points of beginning-of-quarter assets, minus the mean flow of other funds with the same benchmark, averaged over the four quarters before the appointment quarter and across previous funds. Because flows follow returns, this regression controls for the previous funds' benchmark-relative NAV return, $R_h^{\text{NAV,pre}}$, in place of $\mathbf{B}_h$; it uses the 1,322 experienced appointments and otherwise follows Equation (11).[22]

The foreign-education coefficient is −1.8 percentage points per quarter, significant at 10%: relative to U.S.-educated recruits, foreign-educated recruits come from funds with weaker fundraising records even after accounting for prior returns. We also examine subsequent flows to funds with foreign-educated teams. Conditional on past performance and fund and team characteristics, funds with an entirely foreign-educated team receive monthly net flows 0.21 percentage points higher than funds with an entirely U.S.-educated team, with a

[22]This signal is the mean quarterly relative NAV return in the payoff quarters from five to two quarters before the appointment quarter, in percentage points. It is observed before hiring and does not enter either the main or the peer-adjusted investment-record equation.

standard error of 0.15 (Appendix Table A14). Quarterly flows and expense ratios show no clear difference either (Appendix Table A15). The prior-flow result provides no compensating fundraising strength. The subsequent-flow premium is positive but imprecise, leaving its commercial value unresolved. We next examine the investment rationale: whether those recruits bring stronger records in the countries the hiring fund needs.

5.2 Prior performance in the country needing coverage

The weaker international record raises a specialization question: do foreign-educated recruits bring stronger records than U.S.-educated recruits in the particular market needing expertise? We define needed countries before examining the incoming manager's education: foreign destinations already held without a continuing manager's schooling tie.[23] Both education groups are evaluated in this same set, so their prior stock selection is compared in the same needed market. Here an observation is an appointment h paired with a needed country c. The outcome $\bar{R}^{\text{pre}}_{hc}$ is the average adjusted return that the recruit's previous teams earned on holdings in that country over the four full quarters before appointment. Thus, we evaluate the recruit's prior record in the potential specialty rather than averaging across all non-U.S. countries.[24] We estimate

$$\bar{R}^{\text{pre}}_{hc} = \alpha_{jc} + \lambda_{cy} + \beta_G \mathbb{1}_i^{\text{Foreign}} + \beta_L \mathbb{1}_{ic}^{\text{OwnLink}} + \Gamma' X_h + u_{hc}. \tag{12}$$

Here $\mathbb{1}_{ic}^{\text{OwnLink}}$ equals one if the incoming manager studied in country c. Fund-country effects compare recruits to the same fund for the same needed market, and country-year effects remove conditions common to that market in a year. The coefficient β_G compares a foreign-educated recruit with a U.S.-educated recruit in needed countries where the foreign-educated

[23] Continuing managers are present before and at the end of the appointment quarter. Departures and simultaneous recruits do not supply incumbent coverage. Missing prior holdings remain unclassified.

[24] We average share classes within economic origin-quarter, then quarters and then distinct origins. Every origin fund must have a record in all four quarters. A missing country record remains unavailable rather than becoming zero.

recruit did not study, and β_L is the additional difference in the country where the recruit did study. Their sum, $\beta_G + \beta_L$, therefore compares a recruit educated in the needed country with U.S.-educated recruits in that same country. This is a comparison across recruits; appointment fixed effects would instead ask whether the same recruit performed better in a schooling country than elsewhere. This specialist comparison retains the manager and team controls, including continuing-team unknown education, and conditions on the previous funds' benchmark-relative NAV return in place of the benchmark shares. It therefore compares same-country records conditional on overall prior fund performance. Each appointment's supported countries have total weight one, and standard errors are clustered by hiring fund.

The same-country record is weaker, rather than stronger. For a foreign-educated recruit linked to the needed market, and comparing recruits from funds with similar overall prior returns, the estimated difference from U.S.-educated recruits is −274 annualized basket basis points, with a standard error of 101, significant at 1% (Table 10). Their previous teams underperformed the comparison recruits' teams by about 2.7 percentage points a year on holdings in the very market where the schooling link is relevant. The estimate combines the general foreign-education coefficient (−55) and the own-country increment (−219), using their joint covariance. The comparison uses 9,446 country-appointment records at 1,083 appointments and 257 funds; linked records cover 145 appointments and 85 funds.[25]

This comparison strengthens the hiring argument because it evaluates the proposed compensating skill in the market where it is needed. The weaker general record in Table 9 is accompanied by a weaker same-country record in Table 10, not by superior demonstrated specialist performance. The next test asks whether the local specialty's observed payoff accounts for how careers respond to subsequent fund performance.

[25]The 152 linked records span 22 countries, including 72 in the United Kingdom. Starting with 1,131 appointments with supported pre-hire holdings and hiring-fund/year histories, prior-country availability supplies 11,117 records at 1,105 appointments and 269 funds before singleton fixed-effect groups are dropped. Appointments span 2004Q1–2024Q4 and prior payoffs 2003Q1–2024Q3. No fund supplies more than 5.28% of the combined contrast's identifying variation.

5.3 Careers and performance evaluation

The hiring tests establish differences in the records accepted at entry. The career tests ask the next question: once managers are on a fund's team, do their careers respond differently to the same fund performance? Giving foreign education more weight and realized performance less predicts greater relative retention as returns worsen and a smaller relative gain in new assignments as returns improve. We test these responses among all current managers, rather than only the appointments observed in Section 5.1. Each observation is a manager i on fund j at quarter-end q. We study two outcomes over the following quarter: $\mathbb{1}_{ijq}^{\text{Exit}}$ equals one if manager i departs from fund j between quarter-ends q and $q+1$, and $\mathbb{1}_{ijq}^{\text{NewFund}}$ equals one if the manager receives a new assignment at another fund. We multiply both indicators by 100, so coefficients are in percentage points. The signal R_{jq}^{rel} is the fund's compounded net return over quarter q minus the mean return of other funds sharing its benchmark, also in percentage points. It uses reported whole-fund NAV, covering both U.S. and foreign investments. We estimate one common specification separately for departures and new assignments:

$$Y_{ijq} = \mu_{jq} + \nu_{ij} + \kappa \mathbb{1}_{i}^{\text{Foreign}} \times R_{jq}^{\text{rel}} + \Gamma' X_{ijq} + e_{ijq}. \tag{13}$$

Here Y_{ijq} is either $100 \times \mathbb{1}_{ijq}^{\text{Exit}}$ or $100 \times \mathbb{1}_{ijq}^{\text{NewFund}}$. For colleagues on the same fund, the coefficient κ is how much more a foreign-educated manager's probability of leaving, or of a new assignment, changes than a U.S.-educated colleague's when the fund's return rises by one percentage point; we denote the departure and new-assignment coefficients by κ_{Exit} and κ_{NewFund}. Fund-quarter effects μ_{jq} absorb the common return and fund characteristics shared by colleagues, including the benchmark, mandate and team size. Fund-manager effects ν_{ij} absorb each manager's education and persistent differences in their career prospects at the fund. The coefficient is therefore identified from how colleagues' outcomes diverge as their shared fund return varies.

Controls X_{ijq} include log spell length, log one plus management experience, and interactions of R^{rel} with the female, unknown-gender, CFA and unknown-biography indicators and log experience. Each fund-quarter receives equal total weight, divided equally among its managers, and errors are clustered by economic fund.

Departures indicate whether an existing assignment continues; new assignments indicate additional opportunities. A new assignment is a first appearance on another fund's team in the next quarter, including a new fund launch, and can occur without leaving the original fund. The records track assignment changes without classifying their reasons. The two outcomes distinguish relative retention after weak performance from relative access to new assignments after strong performance.

The departure interaction is $\kappa_{\text{Exit}} = +0.145$ percentage points per return percentage point, significant at 1%, on 154,677 manager-quarters at 1,334 funds (Table 11, columns 1–2). A ten-point deterioration lowers the foreign-minus-U.S. departure gap by 1.45 percentage points: foreign-educated managers become less likely to leave relative to their U.S.-educated teammates. For new assignments, $\kappa_{\text{NewFund}} = -0.107$, significant at 5%; a ten-point improvement lowers the foreign-minus-U.S. assignment-probability gap by 1.07 percentage points. The paired signs describe relative protection as returns worsen and a smaller relative gain as returns improve. Both signs are consistent with the hypothesis in Section 2.2 that qualifications receive more weight and shared fund returns less weight in career evaluation. The next comparison asks whether evaluation based on the local specialty can instead explain them.

A specialist may be evaluated on local performance rather than common fund returns. We therefore add the observed performance of education-country holdings and ask whether the foreign-minus-U.S. response to common returns becomes smaller. This complements the hiring test: the prior-country record assesses demonstrated expertise at entry, while the

local-return control asks whether that specialty's observed payoff accounts for later career differences.[26]

Columns 3–4 add this local performance to Equation (13). The local return R^{loc}_{ijq} is the return on the fund's holdings in a foreign-educated manager's education countries over the same quarter, less those countries' index returns, in percentage points. We add it as $\chi \mathbb{1}^{\text{Observed}}_{ijq} \times R^{\text{loc}}_{ijq}$, where $\mathbb{1}^{\text{Observed}}_{ijq}$ equals one for foreign-educated managers whose local return is observed and zero otherwise, including for all U.S.-educated managers, together with the indicator itself, $\phi \mathbb{1}^{\text{Observed}}_{ijq}$. The coefficient χ measures the association between observed local performance and career outcomes, conditional on shared fund performance. If differences in observed local returns account for the groups' different career responses, conditioning on those returns should move κ_{Exit} and κ_{NewFund} toward zero.[27]

Adding local performance leaves the career pattern largely unchanged. The fund-return interactions are +0.138 for departures and −0.108 for assignments, close to the baseline +0.145 and −0.107, and remain statistically significant. The local-return slopes themselves are small and imprecise: 0.017 percentage points for departures and 0.002 for new assignments, per return percentage point. The stability of the fund-return interactions shows that the measured local payoff does not account for the relative career pattern. This complements the weaker same-country records observed at recruitment.

The personnel evidence links weaker records at entry to differential career responses after appointment. The record gap extends to the needed country, while observed local performance does not account for the different career responses. Thus, demonstrated specialization does not explain either the accepted specialty-record gap or the stability of the career pattern. Prior fundraising records also supply no compensating strength. Along-

[26] Appendix B.11 examines recorded coverage categories separately.

[27] All 154,677 manager-quarters remain; 48,073 at 832 funds have observed local returns, and the added return term is zero elsewhere. The local return weights holdings by initial country capital. The coefficients χ and ϕ are estimated separately for each outcome, and controls, effects, weights and clustering are unchanged.

side lower factor-adjusted fund returns and concentrated local gains, this evidence supports overvaluation of geographic familiarity as an ability signal.[28]

6 Conclusion

Funds with greater foreign-educated representation earn lower factor-adjusted net returns, with a similar shortfall before expenses. Outside the education countries, geographic, language and social ties are less favorable for foreign-educated teams than for U.S.-educated teams; foreign-educated teams themselves select stocks worse in bordering and more socially connected markets. Funds overweight their managers' education countries, but foreign-educated teams show no clear average stock-selection advantage there. Selected education-country positions earn high alphas under a common country mix, but at the country weights funds actually choose they add little to overall portfolio alpha.

Because this investment value is narrow, we examined how funds value the background when hiring and evaluating managers. Foreign-educated recruits arrive with weaker prior team records, including education-country holdings, than U.S.-educated recruits hired by the same fund. Their previous funds also attracted weaker flows. In countries needing coverage, recruits educated there have weaker prior team records even conditional on their previous funds' overall performance. Stronger demonstrated specialist performance thus does not explain the weaker records accepted at entry. Relative to U.S.-educated colleagues, their retention improves as shared fund returns worsen, while their access to new assignments declines as returns improve. These differences persist after accounting for observed education-country performance.

Together with the investment evidence, these hiring and career patterns suggest that funds overvalue geographic familiarity when assessing investment ability.

[28]The personnel tests compare accepted recruits and team-level records. Individual responsibilities, rejected applicants and employers' private information are unobserved. The combined evidence supports an overvaluation interpretation; it does not identify employer beliefs or the causal contribution of hiring to the NAV coefficient.

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Table 1: Summary statistics

	Mean	Median	Std. dev.	*N*
A. Portfolios and team education (fund-quarter)				
Number of countries held	18.99	19.00	8.44	67,380
Number of manager education countries (including U.S.)	1.57	1.00	0.92	63,510
Number of non-U.S. manager education countries	0.72	0.00	0.95	63,510
B. Funds (fund-year)				
Total net assets ($ billions)	1.59	0.23	6.66	21,084
Net expense ratio (%)	1.22	1.17	0.50	19,695
Turnover ratio (%)	64.31	48.00	60.96	18,999
Annual return (%)	8.68	10.86	22.59	18,773
Annualized volatility (%)	16.46	14.81	7.09	18,773
C. Managers (one observation per fund)				
Average manager tenure (years)	5.57	4.93	3.27	1,663
Number of managers	3.12	2.44	2.44	1,663
D. Foreign education in the net-return panel				
Foreign-educated manager share (fund-month)	0.35			206,847
Fund-quarters with ForeignShare = 0 (%)	48.6			68,949
Fund-quarters with partly unknown education (%)	28.3			68,949

Notes: This table reports summary statistics for U.S. actively managed international equity funds (Morningstar), 2000–2024. Panel B summarizes fund characteristics. The NAV regressions use returns and assets of the oldest share class; economic-fund clustering combines identifiers linked by the crosswalk. Of the 2,992 managers with recorded post-secondary education, 26.9% studied only outside the United States. Panel A counts the distinct education countries of a fund's current managers. ForeignShare is the share of current managers with any recorded non-U.S. education among managers whose education is known.

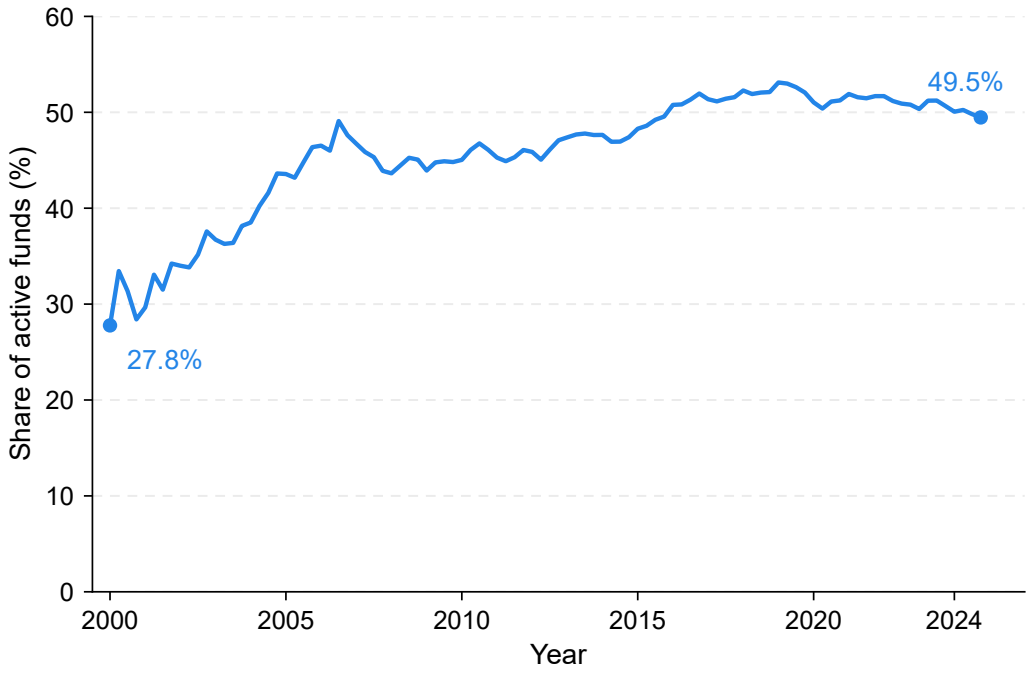


Figure 1: Share of active funds with foreign-educated managers

Notes: Quarterly percentage of active sample funds with at least one current manager with recorded non-U.S. education, 2000Q1–2024Q4. Active funds have positive reported non-U.S. equity holdings in that quarter; no return-availability or stock-classification requirement is imposed. Each economic fund is counted once, with equal weight. Manager status is measured at quarter-end and includes managers educated both in the United States and abroad. All active funds remain in the denominator; unknown education is not classified as U.S.-only. Changes in education coverage can affect the trend. The endpoints are 85 of 306 funds (27.8%) and 369 of 746 funds (49.5%).

Table 2: Foreign education and net fund returns

Dependent variable	Factor-adjusted net NAV return Annualized bps		
Specification	(1) Full controls	(2) Broader sample	(3) No operating controls
Foreign-educated share	−69.51**	−69.30*	−68.31*
	(35.42)	(35.43)	(35.61)
Log team size	−88.89	−87.85	−90.36
	(59.10)	(59.06)	(59.22)
Unknown-education share	−108.22	−108.32	−100.94
	(68.59)	(68.54)	(68.22)
Log lagged assets (USD millions)	−21.83**	−22.38**	−21.14**
	(9.90)	(9.84)	(9.91)
Assets spline basis 1	−100.49***	−99.74***	−99.81***
	(26.04)	(25.98)	(26.16)
Assets spline basis 2	559.63***	556.76***	553.00***
	(158.52)	(158.37)	(158.96)
Team-size spline basis 1	−78.04	−79.95	−77.85
	(137.48)	(137.40)	(137.52)
Team-size spline basis 2	695.54	701.71	686.45
	(590.07)	(589.75)	(589.82)
Female share (known gender)	−50.70	−50.08	−50.77
	(38.54)	(38.52)	(38.58)
Unknown-gender share	−89.18	−90.07	−93.77
	(93.07)	(93.05)	(92.41)
CFA share (observed biographies)	6.58	6.77	7.99
	(31.47)	(31.47)	(31.48)
Unknown-biography share	227.18**	228.44**	227.79**
	(107.77)	(107.71)	(107.52)
Mean log(1 + experience)	−34.70**	−34.15**	−35.16**
	(16.75)	(16.73)	(16.88)
Prior-year relative NAV return (pp)	−4.60***	−4.77***	−4.63***
	(1.46)	(1.46)	(1.47)
Log(1 + prior relative volatility)	62.22**	63.78**	63.77**
	(31.00)	(30.88)	(31.67)
Prior-quarter flow (pp)	1.57×10^{-6}	1.54×10^{-6}	1.55×10^{-6}
	(2.25×10^{-6})	(2.24×10^{-6})	(2.24×10^{-6})
Missing prior return history	99.32	103.71*	75.70
	(62.86)	(62.40)	(57.18)
Missing prior-quarter flow	−41.44	−38.72	−43.26
	(55.66)	(53.41)	(55.30)
Lagged expense ratio (bps)	−0.40	−0.39	—
	(0.34)	(0.34)	
Lagged log turnover	84.01**	82.51**	—
	(35.50)	(35.41)	
Top-ten equity concentration	75.23	68.69	—
	(138.23)	(137.84)	
Missing lagged expense ratio	−111.04**	−111.35**	—
	(54.10)	(53.80)	
Missing lagged turnover	72.29**	71.47*	—
	(36.75)	(36.70)	
Missing equity concentration	26.85	25.74	—
	(42.14)	(42.13)	
Fund FE	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes
Observations	206,847	208,425	206,847
Fund clusters	1,418	1,536	1,418

Notes: Monthly net NAV returns use thirteen predetermined factor exposures, January 2000–December 2024. Coefficients are annualized bps; ForeignShare compares foreign-educated shares of one and zero among known-schooling managers. Columns 1–2 use full controls; column 2 retains shorter-history funds using benchmark-peer loadings. Column 3 omits operating-cost controls. Prior-quarter flow enters in raw percentage points with its missing-flow indicator. Raw flows are uncapped and include very large values, so the flow coefficient is small in these units. No outcome is winsorized or zero-filled. Fund-months have equal weight; standard errors are clustered by economic fund. Standard errors are below coefficients in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 3: Fees and the net-return shortfall

Dependent variable	Net NAV	Gross NAV	Net NAV	Gross NAV
	Factor-adjusted returns, annualized bps			
	(1)	(2)	(3)	(4)
Specification	Full controls	Full controls	No operating controls	No operating controls
Foreign-educated share	−70.56**	−71.03**	−69.29*	−70.13**
	(35.42)	(35.40)	(35.62)	(35.57)
Log team size	−98.60*	−102.57*	−99.66*	−104.30*
	(59.39)	(59.30)	(59.51)	(59.39)
Unknown-education share	−110.61	−112.41	−102.99	−105.70
	(68.97)	(69.14)	(68.57)	(68.76)
Log lagged assets (USD millions)	−21.41**	−21.92**	−20.73**	−21.13**
	(10.01)	(10.03)	(10.03)	(10.04)
Assets spline basis 1	−99.72***	−102.85***	−99.14***	−103.73***
	(26.23)	(26.23)	(26.36)	(26.34)
Assets spline basis 2	551.43***	567.58***	544.25***	564.86***
	(159.79)	(159.72)	(160.21)	(160.07)
Team-size spline basis 1	−53.74	−48.77	−55.10	−49.90
	(137.51)	(137.24)	(137.46)	(137.12)
Team-size spline basis 2	598.98	590.68	597.53	592.21
	(590.13)	(589.18)	(589.31)	(588.13)
Female share (known gender)	−54.16	−54.52	−53.78	−54.23
	(38.51)	(38.51)	(38.57)	(38.56)
Unknown-gender share	−89.22	−89.81	−94.35	−95.36
	(93.32)	(93.38)	(92.67)	(92.86)
CFA share (observed biographies)	6.40	5.06	8.11	6.54
	(31.83)	(31.88)	(31.86)	(31.91)
Unknown-biography share	243.98**	248.18**	245.10**	250.14**
	(108.10)	(108.14)	(107.86)	(107.95)
Mean log(1 + experience)	−34.30**	−35.04**	−34.56**	−35.64**
	(16.81)	(16.80)	(16.93)	(16.92)
Prior-year relative NAV return (pp)	−4.46***	−4.50***	−4.48***	−4.50***
	(1.48)	(1.48)	(1.48)	(1.48)
Log(1 + prior relative volatility)	54.63*	55.14*	56.26*	57.68*
	(31.13)	(31.17)	(31.82)	(31.85)
Prior-quarter flow (pp)	1.43×10^{-6}	1.47×10^{-6}	1.40×10^{-6}	1.47×10^{-6}
	(2.25×10^{-6})	(2.26×10^{-6})	(2.24×10^{-6})	(2.25×10^{-6})
Missing prior return history	107.10*	104.68	78.87	78.88
	(63.81)	(63.90)	(57.63)	(57.65)
Missing prior-quarter flow	−67.65	−68.84	−69.12	−70.44
	(55.95)	(56.00)	(55.76)	(55.81)
Lagged expense ratio (bps)	−0.36	−0.07	—	—
	(0.34)	(0.35)		
Lagged log turnover	83.38**	82.62**	—	—
	(35.68)	(35.65)		
Top-ten equity concentration	71.59	67.49	—	—
	(139.26)	(139.32)		
Missing lagged expense ratio	−120.73**	−84.63	—	—
	(53.46)	(54.45)		
Missing lagged turnover	80.45**	80.34**	—	—
	(35.65)	(35.75)		
Missing equity concentration	32.49	30.96	—	—
	(42.37)	(42.49)		
Fund FE	Yes	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes	Yes
Observations	205,773	205,773	205,773	205,773
Fund clusters	1,418	1,418	1,418	1,418

Notes: Net and reported gross NAV use identical observations and thirteen predetermined factor exposures, 2000–2024. Columns 1–2 use full controls; 3–4 omit expense, turnover and concentration controls. Equal monthly weights; raw prior flow with a missing indicator; economic-fund clustering. Standard errors are below coefficients in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Gross-minus-net foreign-share differences are −0.47 (SE 1.01) for columns 2 minus 1 and −0.84 (SE 1.33) for columns 4 minus 3; paired fund-influence covariance is used.

Table 4: Relatedness and stock selection outside education countries

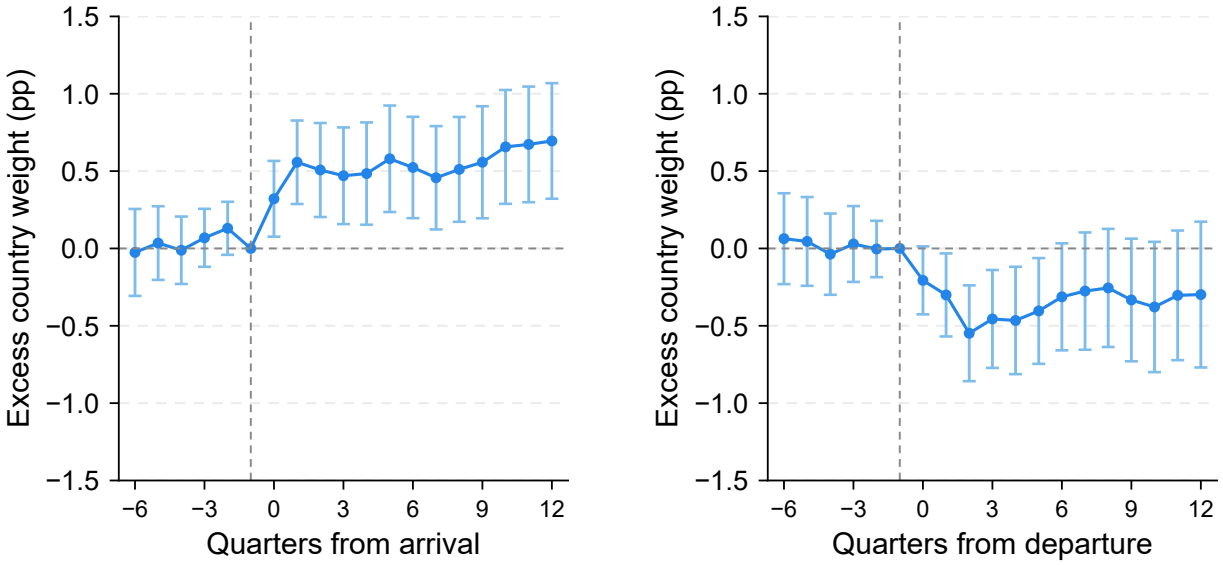
Dependent variable	Adjusted country return Annualized bps		
Specification	(1) Shared border	(2) Official language	(3) Social connectedness
Foreign education × tie (θ)	−484.20***	−230.69***	−109.98***
	(158.36)	(67.31)	(42.05)
Tie (δ)	299.90**	155.35**	67.17
	(134.65)	(75.45)	(43.53)
Foreign-educated share (γ)	−28.66	1.14	826.14**
	(32.33)	(33.18)	(330.63)
Missing-tie share (MissingShare)	8.29	38.62	−333.29*
	(45.48)	(45.98)	(174.15)
Log country stock count	−62.09***	−62.27***	−65.81***
	(11.73)	(11.73)	(11.60)
Log team size	−66.44***	−67.15***	−60.96***
	(21.32)	(21.32)	(22.16)
Unknown-education share	59.19	60.97	43.81
	(84.35)	(85.02)	(83.59)
Female share (known gender)	−95.46**	−96.27**	−103.92**
	(45.61)	(45.61)	(47.46)
Unknown-gender share	54.04	47.36	61.07
	(69.76)	(69.79)	(71.85)
CFA share (observed biographies)	43.11	42.28	44.36
	(30.85)	(30.91)	(31.05)
Unknown-biography share	22.70	30.46	7.04
	(119.49)	(119.55)	(114.17)
Mean log(1 + experience)	−44.05***	−44.47***	−38.68**
	(15.57)	(15.60)	(16.04)
Fund × country FE	Yes	Yes	Yes
Country × quarter FE	Yes	Yes	Yes
Observations	936,076	936,076	876,532
Fund clusters	1,423	1,423	1,421

Notes: Equation (4); the tie averages are defined in Section 4.2. Each column uses one familiarity measure. Column 1 is a shared border; column 2 is a common official language; column 3 is log social connectedness from the October 2021 snapshot, without standardization or centering. Its education-share and missing-share coefficients refer to log connectedness zero; at mean log connectedness 8.107, the comparable education-share level is −65.38 bps. The text translates the raw gradient into a standard-deviation comparison. Foreign-educated managers' ties use their non-U.S. schooling countries; U.S.-educated managers' ties use the United States. Rows report the coefficients of actual regressors in Equation (4). The first row is θ, the coefficient on ForeignTie: the team average of each manager's foreign-education indicator interacted with that manager's tie, not foreign share times the team-average tie. It directly tests the foreign-minus-U.S. difference in slopes. The tie coefficient δ gives the U.S.-educated slope; the foreign-educated slope is their sum. All direct schooling destinations are excluded. Countries receive equal weight within fund-quarter. Standard errors are clustered by economic fund. Samples contain 1,424/1,422 ISINs and 1,423/1,421 economic fund clusters for discrete/SCI measures. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 5: Education links and excess country weights

Dependent variable	Country overweight Percentage points			
	(1)	(2)	(3)	(4)
Specification	Any link	Link share	Any link	Link share
Any manager educated in country	1.108***	—	1.091***	—
	(0.220)		(0.219)	
Share of managers educated in country	—	2.626***	—	2.590***
		(0.459)		(0.454)
Female share (known gender)	0.019	0.025	—	—
	(0.043)	(0.042)		
Unknown-gender share	0.087	0.085	—	—
	(0.060)	(0.059)		
CFA share (observed biographies)	−0.030	−0.032	—	—
	(0.032)	(0.032)		
Unknown-biography share	−0.116	−0.122	—	—
	(0.080)	(0.080)		
Mean log(1 + experience)	0.014	0.013	—	—
	(0.014)	(0.014)		
Missing team-trait record	−0.060	−0.067	—	—
	(0.103)	(0.103)		
Fund FE	Yes	Yes	No	No
Country × month FE	Yes	Yes	Yes	Yes
Fund × month FE	No	No	Yes	Yes
Observations	5,613,030	5,613,030	5,611,449	5,611,449
Fund clusters	1,663	1,663	1,661	1,661

Notes: The outcome is country weight minus benchmark weight, in pp, January 1999–December 2024. Any link is an indicator; link share is the fraction of managers educated in the country. Fund-month effects in columns 3–4 absorb team traits, shown as dashes rather than zeros. Observations have equal weight. Standard errors are clustered by fund identifier. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.



(a) Arrival of the first education-linked manager

(b) Departure of the last education-linked manager

Figure 2: Excess country weight around manager arrivals and departures

Notes: This figure plots event-study coefficients for the fund's country weight minus the benchmark's country weight (percentage points) around the quarter in which the fund gains its first manager with non-U.S. education in the country (left) or loses the last such manager (right). Windows run up to twelve quarters and end if the status reverses; events without a six-quarter clean pre-period and overlapping windows are removed (977 arrivals and 793 departures). Regressions include fund × country, country × quarter and fund × quarter fixed effects, with quarter −1 as the reference. Bars are 95% confidence intervals with standard errors clustered by fund.

Table 6: Stock selection in education and outside countries

Dependent variable	Adjusted country return Annualized bps	
	(1)	(2)
Specification	Equal country weights	Precision weights
Foreign-educated share (outside gap)	−46.31	−31.20
	(31.35)	(28.87)
Education-country link (local increment)	46.52	36.77
	(49.90)	(47.44)
Log country stock count	−57.23***	−52.91***
	(11.42)	(11.36)
Log team size	−67.97***	−38.56*
	(21.46)	(19.84)
Unknown-education share	25.76	2.85
	(84.74)	(78.94)
Female share (known gender)	−96.56**	−117.46***
	(44.95)	(37.52)
Unknown-gender share	94.11	132.75
	(86.85)	(81.54)
CFA share (observed biographies)	51.48*	47.34
	(30.20)	(29.64)
Unknown-biography share	−39.73	−45.65
	(123.43)	(115.07)
Mean log(1 + experience)	−52.48***	−43.23***
	(15.43)	(15.14)
Fund × country FE	Yes	Yes
Country × quarter FE	Yes	Yes
Observations	974,924	974,924
Fund clusters	1,428	1,428

Notes: Equation (7) uses all held foreign countries and the same size/momentum-adjusted next-three-month return. $\mathbb{1}^{\text{Link}} = 1$ if a current manager studied in the country; ForeignShare is the fraction of known-education managers with foreign schooling. The linked-country comparison sets $\mathbb{1}^{\text{Link}} = 1, \text{ForeignShare} = 1$ against $\mathbb{1}^{\text{Link}} = 0, \text{ForeignShare} = 0$. The $\mathbb{1}^{\text{Link}}$ slope alone is not that comparison. Countries have equal weight within each fund-quarter; WLS also weights by inverse fund residual variance. Outcomes cover January 2000–December 2024; 1,429 identifiers map to 1,428 economic-fund clusters. Standard errors are below coefficients in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The linked-country comparisons $\beta + \gamma$ are 0.21 (SE 56.61) and 5.57 (51.39) bps in columns 1 and 2, using joint covariance.

Table 7: Local expertise and high-conviction stock selection

Dependent variable	Portfolio excess return			
	Annual alpha in bps; factor loadings dimensionless			
	(1)	(2)	(3)	(4)
Specification	Local top	Local ordinary	Outside top	Outside ordinary
Alpha (annualized bps)	400.2^{***}	40.1	13.4	38.1
	(133.3)	(95.9)	(54.8)	(39.9)
Global ex-U.S. benchmark excess return	0.148	0.135^{*}	0.208^{***}	0.183^{***}
	(0.106)	(0.076)	(0.043)	(0.032)
Developed ex-U.S. market	0.778^{***}	0.723^{***}	0.638^{***}	0.647^{***}
	(0.097)	(0.069)	(0.040)	(0.029)
Developed ex-U.S. size	−0.012	-0.146^{***}	0.059^{**}	-0.048^{***}
	(0.059)	(0.043)	(0.024)	(0.018)
Developed ex-U.S. value	−0.011	0.021	0.063^{**}	−0.013
	(0.068)	(0.049)	(0.028)	(0.020)
Developed ex-U.S. investment	0.030	−0.059	-0.080^{**}	−0.029
	(0.093)	(0.067)	(0.038)	(0.028)
Developed ex-U.S. profitability	0.015	−0.057	0.055	−0.008
	(0.093)	(0.067)	(0.038)	(0.028)
Emerging-market market	0.103^{**}	0.175^{***}	0.192^{***}	0.183^{***}
	(0.044)	(0.031)	(0.018)	(0.013)
Emerging-market size	0.000	−0.029	0.023	-0.040^{**}
	(0.064)	(0.046)	(0.026)	(0.019)
Emerging-market value	−0.011	−0.019	-0.065^{**}	-0.049^{**}
	(0.066)	(0.047)	(0.027)	(0.020)
Emerging-market investment	−0.041	0.001	0.129^{***}	−0.020
	(0.077)	(0.055)	(0.032)	(0.023)
Emerging-market profitability	−0.064	0.041	0.073^{**}	0.072^{***}
	(0.085)	(0.061)	(0.035)	(0.025)
Observations	300	300	300	300

Notes: Quarter-end active stock weights identify top-quartile and ordinary holdings. Portfolios are value-weighted and reweighted to benchmark country shares, January 2000–December 2024. Annual alpha is $12 \times 10^4 \alpha_p$ basis points; factor loadings are dimensionless. Individual regressions use conventional OLS standard errors. Direct alpha differences and Wald tests use the robust stacked-system covariance without calendar-month clustering. Benchmark annual excess return is 4.30%. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The local top-minus-ordinary alpha is 360.1 bps (SE 137.1, $p = 0.009$); the outside counterpart is −24.7 (SE 60.3, $p = 0.682$). Local-minus-outside equality tests give $p = 0.002$ for top and $p = 0.981$ for ordinary portfolios. Figure 3 displays these four tests.

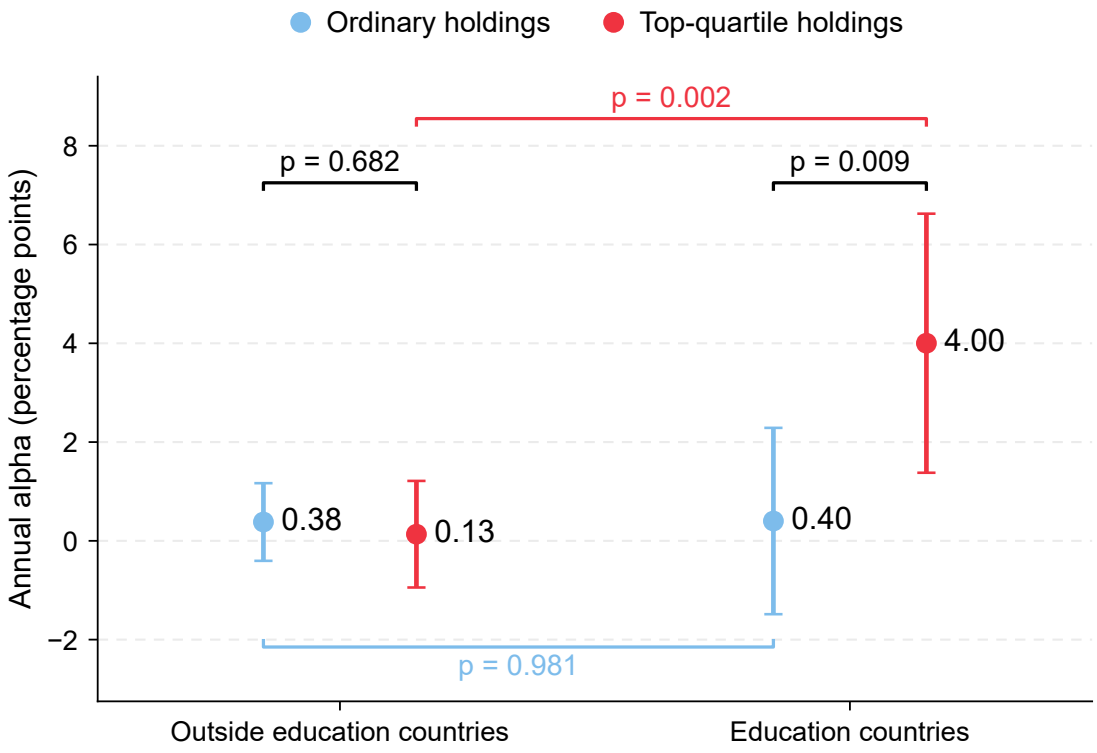
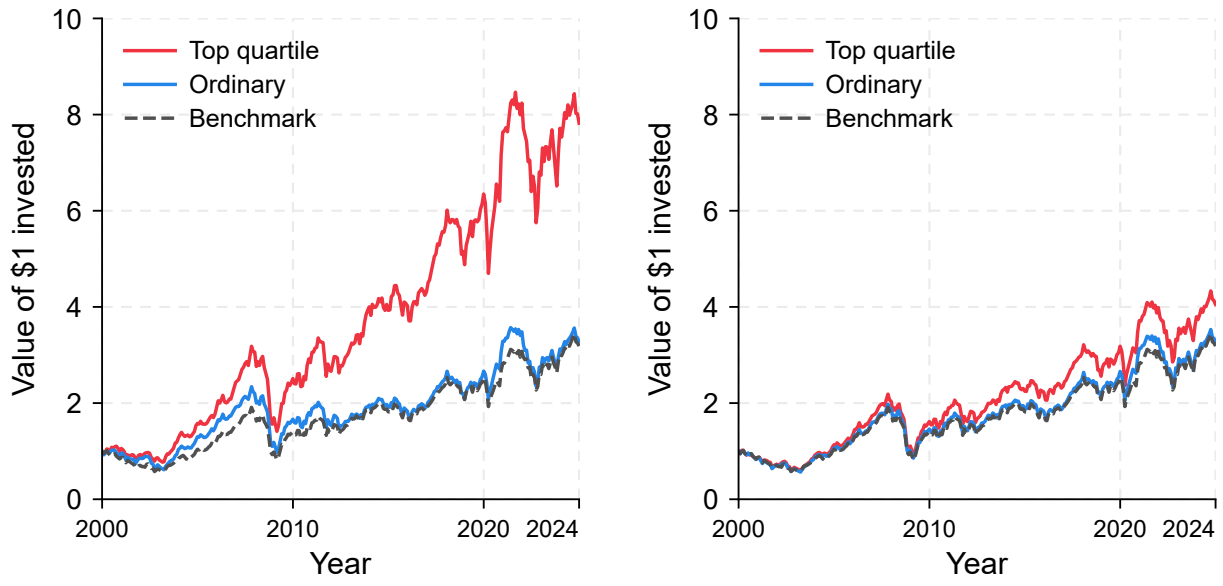


Figure 3: Conviction sorting: portfolio alphas and direct comparisons

Notes: The figure displays the four regressions in Table 7, using January 2000–December 2024. Alpha and its 95% confidence interval are annualized percentage points. Top denotes the highest within-country I-measure quartile; ordinary denotes the other three quartiles. The brackets report all four Wald tests: top versus ordinary within each country group, and education versus outside countries within each conviction group. Individual intervals use conventional OLS inference; differences use a robust stacked covariance. Country portfolios are value-weighted and reweighted to the common benchmark country mix.



(a) Education countries

(b) Outside education countries

Figure 4: Cumulative returns of conviction-sorted portfolios

Notes: Each panel compounds the monthly gross returns of the same top and ordinary strategies used in Table 7. One dollar is invested immediately before January 2000; all 300 monthly returns through December 2024 are included. The dashed line is the common Morningstar Global ex-U.S. benchmark. These are returns on country-reweighted disclosed-holdings strategies, not actual fund NAV, capital contributions or cumulative risk-adjusted alpha. The two panels use the same vertical scale.

Table 8: Actual equity-capital contributions of education-country holdings

Dependent variable	Excess-return contribution		
	Annual alpha in bps; factor loadings dimensionless		
	(1)	(2)	(3)
Specification	All local	Top local	Ordinary local
Alpha (annualized bps)	−9.8	0.3	−10.1*
	(10.8)	(6.1)	(5.8)
Global ex-U.S. benchmark excess return	0.015*	0.008	0.007
	(0.009)	(0.005)	(0.005)
Developed ex-U.S. market	0.078***	0.035***	0.043***
	(0.008)	(0.004)	(0.004)
Developed ex-U.S. size	−0.018***	−0.005*	−0.013***
	(0.005)	(0.003)	(0.003)
Developed ex-U.S. value	0.021***	0.008***	0.013***
	(0.006)	(0.003)	(0.003)
Developed ex-U.S. investment	−0.013*	−0.003	−0.010**
	(0.008)	(0.004)	(0.004)
Developed ex-U.S. profitability	0.029***	0.013***	0.015***
	(0.007)	(0.004)	(0.004)
Emerging-market market	−0.001	−0.004**	0.003*
	(0.004)	(0.002)	(0.002)
Emerging-market size	0.014***	0.006**	0.008***
	(0.005)	(0.003)	(0.003)
Emerging-market value	0.008	0.000	0.007**
	(0.005)	(0.003)	(0.003)
Emerging-market investment	−0.007	0.000	−0.007**
	(0.006)	(0.004)	(0.003)
Emerging-market profitability	−0.006	−0.003	−0.003
	(0.007)	(0.004)	(0.004)
Observations	300	300	300

Notes: Columns 1–3 report capital-weighted excess contributions per dollar of covered disclosed equity, including U.S. equities. Local capital shares average 8.71% overall, 3.62% top and 5.10% ordinary; raw annual contributions are 57.5, 27.3 and 30.1 bps, respectively. All use 300 months, January 2000–December 2024. Loadings are dimensionless; alphas are annualized bps with conventional OLS standard errors. Actual country weights and changing monthly capital shares differ from the benchmark country mix used for the normalized strategies in Table 7. These holdings outcomes are distinct from NAV. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 9: Prior records at appointment

Dependent variable	Prior adjusted return Annualized basket bps (1)	Prior fund flow Quarterly pp (2)
Foreign-educated appointee	−86.41**	−1.83*
	(38.18)	(1.06)
Log pre-hire destination team size	6.17	0.30
	(65.17)	(1.02)
Pre-hire destination unknown education	246.69	1.10
	(172.46)	(2.91)
Log origin team size	−13.50	−2.83**
	(49.99)	(1.18)
Origin-team unknown education	66.20	−7.84
	(287.65)	(7.12)
Log origin country stock count	−80.15	−1.47
	(63.12)	(1.14)
Appointee: top-100 school	54.42	−0.89
	(35.41)	(0.73)
Appointee: school ranked 101–500	106.35**	0.10
	(50.63)	(1.10)
Appointee: female	−21.00	0.83
	(33.75)	(0.85)
Appointee: unknown gender	58.50	1.30
	(85.40)	(2.53)
Appointee: CFA	−56.18*	1.91**
	(31.32)	(0.78)
Log(1 + manager experience)	50.49*	−1.89***
	(29.18)	(0.60)
Prior origin relative NAV (pp)	—	1.73***
		(0.47)
Prior benchmark: China	238.93	—
	(246.40)	
Prior benchmark: Developed Asia-Pacific	−1851.57*	—
	(969.62)	
Prior benchmark: Developed Asia-Pacific ex Japan	238.91*	—
	(139.68)	
Prior benchmark: Developed Europe	569.03***	—
	(193.48)	
Prior benchmark: Emerging Americas	−358.95**	—
	(176.43)	
Prior benchmark: Emerging markets	34.22	—
	(104.25)	
Prior benchmark: Global growth	336.90*	—
	(192.76)	
Prior benchmark: Global large cap	−55.15	—
	(242.52)	
Prior benchmark: Global	−82.44	—
	(127.64)	
Prior benchmark: Global small/mid cap	14.81	—
	(187.87)	
Prior benchmark: Global value	−107.66	—
	(228.04)	
Prior benchmark: Global ex-U.S. growth	7.68	—
	(120.20)	
Prior benchmark: Global ex-U.S. small/mid cap	180.53	—
	(135.85)	
Prior benchmark: Global ex-U.S. value	−170.82	—
	(112.42)	
Prior benchmark: Japan	3214.77	—
	(2386.77)	
Hiring-fund FE	Yes	Yes
Appointment-year FE	Yes	Yes
Observations	1,321	1,322
Hiring-fund clusters	300	300

Notes: Equation (11). Reference group: U.S.-educated appointees. Returns include all supported non-U.S. countries with size/past-return adjustment and equal country weights. Flows are benchmark-relative. Omitted benchmark: Global ex-U.S. Equal appointment weights; hiring-fund-clustered standard errors in parentheses. *, ** and *** indicate 10%, 5% and 1% two-sided significance.

Table 10: Demonstrated performance in countries needing coverage

Dependent variable	Prior adjusted return in needed country Annualized basket bps (1)
Own schooling link to needed country	−218.86**
	(103.19)
Foreign-educated appointee	−55.14
	(35.89)
Log pre-hire destination team size	9.54
	(74.69)
Pre-hire destination unknown education	1035.08***
	(257.23)
Log origin team size	1.43
	(55.61)
Origin-team unknown education	113.97
	(254.52)
Log origin country stock count	−193.92**
	(76.70)
Appointee: top-100 school	−46.52
	(34.20)
Appointee: school ranked 101–500	6.88
	(47.37)
Appointee: female	55.12
	(44.47)
Appointee: unknown gender	−23.44
	(126.33)
Appointee: CFA	−21.10
	(32.48)
Log(1 + manager experience)	57.99
	(36.41)
Prior origin relative NAV (pp)	245.86***
	(22.87)
Continuing-team unknown education	−582.72**
	(251.81)
Hiring-fund × country FE	Yes
Country × appointment-year FE	Yes
Country-appointment observations	9,446
Appointments	1,083
Hiring-fund clusters	257

Notes: Equation (12). The outcome is prior team performance in the same needed country over four full pre-hire quarters. The own-schooling link identifies the recruit's exact education country. The linked foreign-minus-U.S. comparison is −274.00 (SE 100.61), the sum of the two focal slopes with joint covariance. This model conditions on prior overall NAV performance. Each appointment has equal total weight across supported countries. The 152 linked observations cover 145 appointments at 85 funds. Hiring-fund-clustered standard errors are below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table 11: Career evaluation and local performance

Dependent variable	Departure	New assignment	Departure	New assignment
			Probability in pp	
	(1)	(2)	(3)	(4)
Specification	Baseline	Baseline	With local performance	With local performance
Foreign-educated × fund return (pp)	0.145^{***}	-0.107^{**}	0.138^{***}	-0.108^{**}
	(0.044)	(0.053)	(0.044)	(0.054)
Observed education-country holdings	—	—	−0.17	0.65^{*}
			(0.35)	(0.39)
Local holdings × local return (pp)	—	—	0.017	0.002
			(0.013)	(0.016)
Log recorded fund spell length	2.70^{***}	-2.43^{***}	2.71^{***}	-2.43^{***}
	(0.28)	(0.35)	(0.28)	(0.35)
Female × fund return (pp)	0.071	−0.029	0.071	−0.030
	(0.056)	(0.076)	(0.056)	(0.076)
Unknown gender × fund return (pp)	0.006	0.468^{**}	0.007	0.469^{**}
	(0.139)	(0.210)	(0.139)	(0.210)
CFA × fund return (pp)	−0.050	−0.044	−0.050	−0.044
	(0.043)	(0.050)	(0.043)	(0.050)
Unknown biography × fund return (pp)	−0.107	−0.577	−0.109	−0.580
	(0.248)	(0.380)	(0.249)	(0.380)
Log(1 + experience) × fund return (pp)	0.045	−0.027	0.045	−0.027
	(0.029)	(0.036)	(0.029)	(0.036)
Log(1 + manager experience)	-3.15^{***}	1.74^{***}	-3.15^{***}	1.73^{***}
	(0.41)	(0.56)	(0.41)	(0.56)
Fund × quarter FE	Yes	Yes	Yes	Yes
Fund × manager FE	Yes	Yes	Yes	Yes
Observations	154,677	154,677	154,677	154,677
Fund clusters	1,334	1,334	1,334	1,334

Notes: All columns estimate Equation (13) separately for departures and new assignments. Columns 3–4 add the observed-local-holdings indicator and its interaction with local performance. Outcomes are next-quarter departures and new assignments, in percentage points. Prior-quarter benchmark-relative fund return and local return are in percentage points; interaction coefficients are percentage points per return percentage point. The foreign interaction is a difference in performance slopes relative to U.S.-educated teammates, not an absolute exit probability. Managers receive inverse team-size weights so each fund-quarter has equal total weight. Standard errors are clustered by economic fund. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

A Samples and descriptive statistics

Tables A1–A5 describe the funds, outcome-specific samples, mandates, holdings and recorded schooling links. Differences in estimation coverage reflect the data needed for each comparison rather than changes to the set of funds.

A.1 The within-fund appointment trend

Figure 1 describes the prevalence of foreign-educated managers on current fund teams. Here we examine new appointments. An appointment is a new manager spell at an existing observed team; initial appearances at a fund's launch or the beginning of its history are excluded. The 6,303 known-education appointments include assignments within fund families and moves between firms. The foreign-educated share rises from 32.3% in 2000–2004 to 42.5% in 2020–2024. This supplementary regression asks whether the rise also occurs within funds, rather than reflecting only changes in which funds appoint managers.

To distinguish a rise within funds from a change in the mix of funds, we estimate

$$H_{jq} = \alpha_j + \tau t_q + \Gamma' X_{jq} + e_{jq}. \tag{14}$$

Here H_{jq} is the percentage of known-education appointees who are foreign-educated, measured from 0 to 100. Calendar time t_q is the year of appointment quarter q minus 2012. Controls X contain the size and five traits of the hiring fund's team before the appointments and the unknown-education fraction among new appointees. Observations are fund-quarters with known-education appointments, weighted by their number, with fund effects and fund-clustered errors. The focal coefficient τ is the within-fund appointment-share change per year: +0.60 percentage points per year (+6.0 per decade), significant at 1%, on 4,072 fund-quarters at 1,218 funds (Table A6). Thus, the increasing foreign-educated appointment share is also present within hiring funds.

B Supplementary results

The appendix follows the paper's evidence chain. We first assess the NAV comparison's weighting, controls and appointment timing, then the time profile of the outside-country comparison and the familiarity results with raw country returns. We then examine a possible commercial payoff, the comparability of recruits' records, and the role of country coverage in careers. Finally, local-portfolio checks assess how narrowly the positive investment application is concentrated. Panel models use economic-fund clustering; portfolio alphas use the time-series inference in Section 4.5.

B.1 Country relevance measured before appointment

This extension of Equation (3) tests whether the performance association is more favorable when managers' schooling countries have greater benchmark relevance at appointment. Let a_{ih} denote benchmark capital in manager i's distinct non-U.S. education countries, averaged over the four quarter-ends strictly before appointment h. Schooling includes undergraduate and graduate degrees. The denominator is total benchmark capital, including U.S. equities, and multiple degrees in the same country count once. All four quarters must have at least 95% of benchmark weight mapped to countries; a 105% upper bound allows rounding. Missing pre-appointment coverage is classified as unknown. The recorded benchmark identifier is static, so a_{ih} uses historical weights of that index rather than a verified historical mandate.

The added regressor Relevance_{jq} is the sum of measured a_{ih} across current foreign-educated managers divided by the number of current managers with known education; unmeasured contributions are set to zero and identified separately by $\text{UnknownRelevance}_{jq}$, the share foreign-educated with unknown relevance. We add $\theta\text{Relevance}_{j,q-1}$ and $\eta\text{UnknownRelevance}_{j,q-1}$ as separate terms to Equation (3), with its existing $\rho\text{ForeignShare}_{j,q-1}$ retained. Unknown education remains a distinct control. Relevance is fixed within each appointment spell; changes in the current team's membership and composition generate variation within funds. No fund-month is dropped for missing relevance, which affects 50.4% of foreign manager-quarter observations.

Table A7, column 2, also includes foreign education-country shares. A manager with K distinct foreign education countries contributes $1/K$ to each, aggregated over the same team denominator. Their sum is ForeignShare, so the UK share is omitted; Lebanon is absorbed by the fixed effects. Accordingly, ρ in this column is conditional on the omitted country and zero measured relevance, not the main paper's average foreign-share comparison. The coefficient of interest is θ. Its raw units are annual NAV basis points per unit of team relevance. The interquartile contrast is 0.1050θ for a fully foreign-educated team with known relevance: -48.6 in column 1 and -72.3 in column 2, with 95% intervals $[-109.4, 12.2]$ and $[-122.7, -21.9]$. At a smaller fixed foreign share, the contrast scales with that share if known-relevance managers' exposure changes by the same amount. Using current benchmark weights instead of pre-appointment weights gives imprecise gradients with and without country-composition controls; that alternative also changes which managers have measured relevance. The adverse association reported here concerns relevance recorded at appointment.

B.2 Precision weighting

Table A8 asks whether less noisy funds show a similar performance gap. It retains the factor-adjusted net NAV outcome, observations, controls and fixed effects of Table 2, column 1, and weights observations by the inverse of the fund's residual return variance. The foreign-education coefficient is -69.7 annual basis points (standard error 22.7), close to the equally weighted estimate. The baseline is reported once, in the main table.

B.3 Further net-return specifications

Quarterly flow is $\text{Flow}_{jq} = 100[\text{TNA}_{jq} - \text{TNA}_{j,q-1}(1 + R^{\text{net}}_{jq})]/\text{TNA}_{j,q-1}$, where TNA is quarter-end assets and R^{net}_{jq} the compounded quarterly net return in decimals. Thus Flow is in percentage points of beginning-quarter assets; the NAV model uses the preceding quarter's $\text{Flow}_{j,q-1}$ for the months of quarter q. On the identical 206,847 fund-months, replacing raw prior-quarter flow with $\text{asinh}(\text{Flow}_{j,q-1}/10)$ changes the foreign-education coefficient from -69.51 to -69.49 annual basis points. The paired raw-minus-transformed difference is -0.02 basis points, with a 95% interval of $[-0.45, 0.42]$. We use raw flow for its simpler interpretation. Both comparisons retain uncapped flow observations; the exercise tests the sensitivity of the education coefficient to the control's transformation.

Table A9 retains Equation (3), its sample and fixed effects. Column 1 adds portfolio size and momentum tilts, the holdings-weighted average size and past-return terciles (coded 0 to 2) of the fund's foreign stocks within the country groups that define the stock benchmarks in Section 3.3, to the model without operating controls; column 2 instead adds top-ten concentration and turnover. Their missing-information indicators retain unsupported controls. Column 3 uses the full controls and winsorizes the risk-adjusted outcome at the 1st and 99th percentiles within benchmark-month cells. All use equal fund-month weights and economic-fund clustering. The focal education-share

coefficients are −66, −69 and −69 annual basis points. These comparisons assess the stability of the return gap to measured portfolio composition and extreme outcomes.

B.4 Staffing timing and selection sensitivity

Two diagnostics examine staffing timing and selection on observed characteristics. If weak current returns forecast a future shift toward foreign-educated managers, future team composition could help explain the current association. We add future foreign-educated share and future unknown-education share to Equation (3), measured four quarters after the team in Equation (3): $\rho_{+4}\text{ForeignShare}_{j,q+3}$ and $\xi_{+4}\text{UnknownShare}_{j,q+3}$, where q is again the quarter containing month m.

The controls X are those of Equation (3), including the current unknown-education share. The coefficient ρ_{+4} measures the additional predictive content of future foreign-educated share, conditional on the current team and controls. On the 193,986 fund-months with teams observed four quarters ahead, the current-share coefficient is −75 basis points (Table A10, column 1). Including future foreign-educated and unknown-education shares leaves it at −72; the future foreign-educated share has a coefficient of −3 with a standard error of 37 (column 2). The current association is thus stable to future team composition, which adds no clear independent predictive relation.

The second diagnostic asks whether observed fund characteristics account for the negative coefficient. With only the unknown-education fraction controlled, the estimate is −59 basis points; adding the other controls makes it −70 (column 3 and Table 2). Within the proportional-selection calibration of Oster (2019), the adjustment at equal observed and unobserved selection gives −73 basis points, while the selection parameter needed for a zero coefficient is −20.6.[29] The observed-control adjustment moves the coefficient away from zero. Together, these checks document stability to the measured staffing and fund conditions in Equation (3).

B.5 Performance beyond the initial appointment period

To test whether the NAV association is confined to newly appointed managers, Table A11 modifies Equation (3) by replacing $\text{ForeignShare}_{j,q-1}$ with three shares: $\text{ForeignShare}^{\text{new}}_{j,q-1}$, $\text{ForeignShare}^{\text{est}}_{j,q-1}$ and $\text{USShare}^{\text{new}}_{j,q-1}$. Their coefficients are ρ_F^{new}, ρ_F^{est} and ρ_{US}^{new}, respectively.

The three shares count new foreign-educated managers, established foreign-educated managers and new U.S.-educated managers, respectively, divided by all managers with known education. Established U.S.-educated managers form the omitted category. New means within the first two years (eight quarters) at the fund, and established means later; as in the NAV equation, q contains the return month and the shares are measured at quarter-end $q-1$. The reduced controls X contain unknown schooling, team and asset scale with spline terms, and the five team traits. Fund and benchmark-month effects, the predetermined thirteen-factor outcome and fund clustering are retained; OLS and precision-weighted columns use the same model. The focal established-share coefficient ρ_F^{est} is −81 annual basis points in OLS and −82 in WLS, significant at 5% and 1%. The new foreign-minus-U.S. contrast, $\rho_F^{\text{new}} - \rho_{US}^{\text{new}}$, is −35 in OLS and imprecise. The evidence of underperformance therefore extends beyond a temporary arrival period.

[29] We set $R_{\max} = 1.3$ times the within-fund R^2 of the controlled regression. Within explanatory power is low for monthly returns, so the calibration is driven substantially by the direction of coefficient movement. It is a sensitivity calculation under proportional selection, not a measured amount of unobserved confounding.

B.6 Changes in the geographic performance gap

We next ask whether the foreign-versus-U.S. comparison in outside countries changes over the twenty-five-year sample. To make the local and outside comparisons consistent, we retain the full country panel in Equation (7) and allow both education associations to vary with calendar time:

$$\begin{aligned} R_{jcq} = \alpha_{jc} + \lambda_{cq} + \beta \mathbb{1}^{\text{Link}}_{jcq} + \gamma \text{ForeignShare}_{jq} \\ + \beta^T \mathbb{1}^{\text{Link}}_{jcq} \times t_{q+1} + \gamma^T \text{ForeignShare}_{jq} \times t_{q+1} + \Gamma' X_{jcq} + \varepsilon_{jcq}. \end{aligned} \tag{15}$$

As in Equation (14), t_q is the year of quarter q minus 2012, in single years; the superscript T marks the time-interaction coefficients. Holdings and education at quarter-end q predict returns in $q+1$, so the interactions use the payoff year t_{q+1}. The sample, controls, weights, fixed effects and clustering are those of the joint country model in Section 4.4.

The outside-country gap at date t is $g_{\text{out}}(t) = \gamma + \gamma^T t$. For a fully foreign-educated team in an education country, the comparison with a U.S.-educated team is $g_{\text{local}}(t) = \beta + \gamma + \beta^T t + \gamma^T t$. Each uncertainty calculation uses the joint covariance of its components. The endpoint gaps are fitted values from the full-sample trend regression.

The outside-country gap becomes more negative over time. In Table A12, column 1, γ^T is -7.06 annualized basket basis points per calendar year, significant at 5%. The fitted gap reaches -124.5 by 2024, with a standard error of 48.3. Inverse-residual-variance weighting gives a similar deterioration and endpoint gap (column 2). Figure A1 plots the outside and local comparisons from these same regressions. The late-sample outside shortfall is about 1.25 percentage points per dollar of the country basket.

The education-country comparison remains imprecise in 2024 under both weighting schemes. The evidence for local expertise comes from the direct conviction-portfolio contrasts in Section 4.5.

Country-quarter effects remove common national-market returns, and the stock comparisons control size and momentum. The estimate captures the time pattern in stock selection after these adjustments.

B.7 Familiarity results with raw country returns

The adjusted country return in Equation (2) subtracts the return on a matched portfolio of local stocks with similar size and past returns. Table A13 re-estimates Equation (4) with the raw return on the fund's holdings in each country instead. The observations, controls, fixed effects, weights and clustering are those of Table 4. Country-quarter effects remain, so the comparison is still among funds holding the same country in the same quarter; only the matching on stock size and past returns is dropped.

The familiarity pattern remains. The difference between the foreign- and U.S.-educated slopes, θ, is -346 basis points for a shared border (standard error 155, significant at 5%), -195 for a common official language (standard error 69, significant at 1%) and -81 basis points per log point of social connectedness (standard error 41, significant at 5%). The foreign-educated slopes, $\delta + \theta$, are -157 for borders (standard error 74, significant at 5%), -62 for official language (standard error 64) and -35 per log point of social connectedness (standard error 17, significant at 5%). The raw differences are 15 to 30 percent smaller than those in Table 4. For borders, the change comes mainly from U.S.-educated teams' holdings in Canada and Mexico, whose size and past-return characteristics earned lower returns: without the benchmark, the U.S.-educated border slope falls from +300 to +188 basis points, while the foreign-educated slope moves only from -184 to -157.

B.8 Performance-conditioned monthly flows

A commercial payoff could make the background valuable even without stronger investment performance; investor preferences can shape fund incentives (Gantchev et al., 2024). The broad monthly flow test separates a potential commercial value of foreign education from investors' responses to prior returns. Let TNA_{jm} be end-of-month assets and R^{net}_{jm} the monthly net return in decimal units. Flow in percentage points is $\text{Flow}_{jm} = 100[\text{TNA}_{jm} - \text{TNA}_{j,m-1}(1 + R^{\text{net}}_{jm})]/\text{TNA}_{j,m-1}$. Before estimation, this monthly flow measure is winsorized at the pooled 1st and 99th percentiles; the cutoffs appear in Table A14. This treatment applies to the monthly flow outcome, not the raw quarterly flow control in the NAV model. We estimate

$$\begin{aligned}\text{Flow}_{jm} = \alpha_j + \lambda_{bm} + \delta\text{ForeignShare}_{j,q-1} \\ + \Gamma'_S S_{j,m-1} + \Gamma'_X X_{j,m-1} + \Gamma'_Z Z_{j,q-1} + u_{jm}.\end{aligned} \tag{16}$$

$\text{Rank}_{j,m-1}$ is the zero-to-one percentile rank of the twelve-month compounded return ending in $m-1$ among same-benchmark funds. Suppressing those subscripts within the three expressions, $S_{j,m-1}$ contains $\min(\text{Rank}, .2)$, $\min(\max(\text{Rank} - .2, 0), .6)$ and $\max(\text{Rank} - .8, 0)$. These terms allow flows to respond differently to poor, middle and strong prior performance. X contains log lagged assets, log fund age, the standard deviation of the preceding twelve monthly returns, lagged annual expenses and log(1+turnover). Z contains log team size, unknown-education share and the five biography traits. As in Equation (3), b is the assigned benchmark and team measures use the quarter-end $q-1$ preceding the quarter q that contains month m. The observed-flow sample has 185,748 months at 1,494 funds; missing required characteristics are not filled with zero.

The coefficient δ asks whether foreign-educated representation predicts additional flows among funds with comparable prior performance and characteristics. Table A14 gives $\delta = 0.207$ monthly percentage points for the full-share difference, with a standard error of 0.153 and a 95% interval from -0.093 to $+0.508$. Thus a positive commercial premium is possible but not precisely established. The model conditions on prior observed performance and measures subsequent flows associated with team composition. The quarterly and appointment comparisons in Table A15 retain their own outcome definitions and samples. Together these tests compare prior fundraising records with subsequent asset gathering as possible sources of commercial value.

B.9 Commercial outcomes at quarterly and appointment horizons

Table A15 asks whether the qualification brings a commercial payoff despite its weaker investment-return association. The outcomes $Y^{(v)}_{jq}$ follow the team observed at quarter-end q: flow in quarter $q+1$ (columns 1–2), the sum of flows over quarters $q+1$ to $q+4$ (column 3), and the expense ratio reported for the previous year (column 5). Flow is measured in percentage points of beginning-quarter assets and expenses in annual basis points. For each outcome v, we estimate

$$Y^{(v)}_{jq} = \alpha_j + \lambda_{bq} + \delta_v\text{ForeignShare}_{jq} + \Gamma'_v X_{jq} + u^{(v)}_{jq}. \tag{17}$$

Fund effects α_j and benchmark-quarter effects λ_{bq} follow Equation (3). The focal coefficient δ_v compares an entirely foreign-educated team with an entirely U.S.-educated team, conditional on the stated controls. A positive flow coefficient would indicate greater subsequent asset gathering; a positive expense coefficient would indicate a higher reported fee rate. The latter is a comparison of existing fees, not a subsequent fee change. The controls X contain UnknownShare, log team size and lagged assets with their spline terms, and the five team traits. Column 2 adds R^{past}_{jq} and the

interaction $\text{ForeignShare}_{jq} \times R^{\text{past}}_{jq}$ as separate regressors, where R^{past}_{jq} is the prior-year relative NAV return, the same variable as the NAV-model control, in annual percentage points. It differs from the single-quarter career signal R^{rel}. Its interaction measures the foreign-minus-U.S. difference in flow-performance sensitivity.

Column 4 asks the more specific question of whether foreign-educated appointments are followed by stronger fundraising. Its outcome is the sum of flows over the four quarters after appointment. In Equation (17), it replaces current-team ForeignShare_{jq} with the appointee fraction s^H_{jq} and benchmark-quarter effects with appointment-year effects; we denote the appointee-share coefficient by δ_H.

The fraction s^H_{jq} is measured among known-education appointees. It represents the same composition as H_{jq} in Equation (14), expressed from zero to one rather than zero to 100. Thus δ_H compares subsequent four-quarter flows after all-foreign versus all-U.S. appointments, conditional on prior fund conditions. The controls X contain pre-appointment four-quarter flows, the prior-year relative NAV return, lagged assets, the size and traits of the hiring fund's team before the appointment, and unknown education among new appointees. All commercial models use equal observation weights and cluster by economic fund. The full-share differences are -0.29 flow percentage points next quarter, -0.26 over four quarters, and $+0.22$ over four quarters after appointment. The expense difference is -0.39 annual basis points. These estimates do not establish a commercial premium. The flow-performance interaction is -0.04 percentage points per annual NAV percentage point (standard error 0.06), also imprecise; it asks whether investors' flow response to performance differs with the team's education share.

B.10 Prior records relative to benchmark country and quarter peers

Direct benchmark controls in the main hiring comparison allow different mean records across origin mandates. The checks here retain the earlier outside-country outcome. A finer comparison also accounts for the particular countries and quarters in which the previous funds invested. For each origin-country-quarter record, we subtract the mean selection of other economic funds sharing its assigned benchmark, investment country and quarter; all share classes of the focal economic origin are excluded. Peer share classes are combined before averaging economic funds. We then apply the same equal-country, equal-quarter and equal-economic-origin record weights to form $\bar{R}^{\text{peer}}_h$, which replaces $\bar{R}^{\text{pre}}_h$ as the outcome in Equation (11).

The outside-country record and its peer-adjusted counterpart use identical supported countries, 1,321 appointments and 27 regressors. Both retain hiring-fund and year effects, equal appointment weights and fund clustering; neither controls for prior NAV. Benchmarks are supplied assigned mandates, rather than independently reconstructed historical mandate changes.

The peer-adjusted foreign-education coefficient is -71.8 annualized basket basis points, with a standard error of 41.0, significant at 10% (Table A16). The point remains negative after the finer origin-country/time comparison. The corresponding outside-country estimate without peer subtraction is -100.8; the all-country main estimate is -86.4. These comparisons distinguish peer adjustment from country scope.

A separate comparison asks whether the record gap remains among recruits accepted together. Table A17 uses the outside-country version of $\bar{R}^{\text{pre}}_h$ as the outcome in Equation (11), replaces separate fund and year effects with hiring-fund-by-appointment-quarter effects α_{jq}, where q is the appointment quarter, and omits $\mathbf{B}_h$. We denote its foreign-education coefficient by δ_{round}.

Hiring-fund-by-appointment-quarter effects hold common recruitment conditions fixed; destination team size and unknown schooling are absorbed. This model uses all available outside-country records and no origin benchmark or prior NAV control. Equal appointment weights and hiring-fund clustering are retained. On 832 appointees, δ_{round} is -81 annualized basket basis points, with a standard error of 41 ($p = 0.053$). Thus foreign-educated recruits have weaker observed prior team records even within a common appointment round. The negative point remains among appointees accepted together, separately from the origin-mandate adjustments above.

B.11 Country coverage and career evaluation

Table A18 separates the career comparison by schooling coverage: a foreign-educated manager's coverage is unique if a held country is covered by no other current manager's schooling, duplicated if a colleague also studied there, unmatched if no education country is held, and unclassified if prior holdings are missing. These categories are measured each quarter from the previous quarter's holdings and the current team. The equation replaces $\mathbb{1}_i^{\text{Foreign}} \times R_{jq}^{\text{rel}}$ in Equation (13) with category interactions:

$$100 \times \mathbb{1}_{ijq}^{\text{Exit}} = \mu_{jq} + \nu_{ij} + \sum_g \kappa_g \mathbb{1}_{ijq}^g \times R_{jq}^{\text{rel}} + \sum_{g \neq \text{NoMatch}} \eta_g \mathbb{1}_{ijq}^g + \Gamma' X_{ijq} + e_{ijq}. \tag{18}$$

Here g indexes unique, duplicated, no-match and unclassified coverage, and $\mathbb{1}_{ijq}^g$ is the corresponding foreign-educated-manager indicator; U.S.-educated colleagues are the comparison. Because the four indicators sum to $\mathbb{1}_i^{\text{Foreign}}$, which the fund-manager effects absorb, the no-match level is omitted, and all four performance interactions enter alongside the career controls, fund-quarter and fund-manager effects, inverse team-size weights and fund clustering. The focal κ_g coefficients are relative exit-response slopes per percentage point of better performance: +0.180 for unique coverage, +0.006 for duplicated coverage and +0.122 for no held-country match. Multiplying by a ten-point deterioration gives relative departure differences of -1.80, -0.06 and -1.22 percentage points. Unique and no-match slopes individually exclude zero, but direct group differences are imprecise; the joint zero-response test for duplicated and no-match coverage has $p = 0.141$. Relative protection can therefore appear beyond currently matched holdings without precisely distinguishing the coverage groups. Coverage measures recorded schooling in held countries, rather than verified responsibilities or future country plans.

B.12 Alternative local-portfolio constructions

Table A19 uses Equation (9) to ask whether the local high-conviction result depends on its stock-ranking or risk model. Columns 1–4 retain the baseline factors while changing the within-country reference weights, the two-year reference universe, U.K. inclusion or undergraduate-only link definition. Columns 5–6 add developed and emerging momentum, retaining or omitting the common portfolio benchmark. Column 7 retains only that benchmark, and column 8 deducts estimated bid-ask trading costs. Each column reports one top-portfolio regression with all loadings; the table notes report the paired Wald tests of top-minus-ordinary alpha. The baseline is shown only in Table 7. Alphas range from 2.75% to 5.74%. The spread remains significant at 5% for the U.K., undergraduate, momentum, benchmark-only and trading-cost variants; it is marginal for the expanded two-year reference set ($p = 0.057$) and imprecise under benchmark reference weights ($p = 0.251$).

C Local Portfolio Weighting Reconciliation

This reconciliation separates the value of stock choices under a common country mix from their payoff at actual portfolio allocations. Table A20 holds the source countries and months fixed while comparing top, ordinary and combined local portfolios under benchmark and actual country weights. The main contribution test in Table 8 then weights each month's actual local return by its share of covered equity capital. These comparisons concern the constructed local strategies; the joint country panel in Table 6 separately compares teams with U.S.-educated peers.

The country-weighting comparison holds local stock baskets fixed and changes the relative capital assigned to their countries. On 77,188 identical fund-country-month cells at 749 identifiers, actual country weights lower the combined local alpha by 2.3 percentage points per year relative to benchmark weights ($p = 0.007$; Table A20). This is a conditional comparison within the local book. It is not a 2.3-percentage-point loss on total fund capital and cannot be added to the panel's education coefficient.

Table A20 re-estimates Equation (9) for six portfolios on identical source support and months. Columns 1–3 use benchmark country shares for top, ordinary and combined local holdings; columns 4–6 use actual capital weights. Risk factors, the common portfolio benchmark and time-series inference are unchanged. The focal comparison is $12 \times 10^4(\alpha_{\text{actual,all}} - \alpha_{\text{benchmark,all}})$, reported in the table notes: -234 annual basis points, significant at 1%. Actual country deployment captures less alpha than the common-country comparison. The combined portfolios are estimated directly, since a constant average sleeve share cannot reproduce time-varying weights.

D Variable Definitions and Timing

Table A21 defines the main variables. A basis point is one hundredth of a percentage point. Coefficients on ForeignShare compare foreign-educated shares of one and zero among managers with known schooling, holding the unknown-education fraction fixed. Fund returns, adjusted country returns and portfolio alphas are all annualized but have different capital denominators and cannot be added. Missing education is coded as unknown, and an unavailable return is not treated as zero.

Fractions are coded from zero to one unless a percentage-point unit is stated. Return controls and signals use percentage points; capital coverage uses percentage points; calendar interactions use years from the stated reference date. Social connectedness enters in log units without standardization. Economic changes such as ten percentage points or one standard deviation are calculated in the text. Log experience is $\log(1 + \text{years of experience})$; the team measure is its mean. The two nonlinear asset terms are restricted-cubic-spline bases in log lagged assets, with knots $(0.5921, 4.1586, 5.8772, 8.4417)$; the corresponding team-size knots are $\log(1, 2, 3, 5)$. The linear log term is reported separately. The prior relative NAV control in the NAV and quarterly commercial models is the preceding year's annualized monthly mean in percentage points; the monthly flow model instead uses ranks of compounded twelve-month returns. Prior relative volatility is annualized in percentage points before applying $\log(1+\cdot)$. The main and peer-adjusted appointment selection models contain origin benchmark indicators and no prior NAV control. Separate appointment flow and country-coverage models retain a mean pre-appointment quarterly benchmark-relative NAV signal in percentage points; its payoff quarters run from five to two quarters before the appointment quarter. Prior flow enters directly in percentage points of beginning-of-quarter assets, with a missing-information indicator. Separate missing-information indicators retain unavailable controls. Fixed-effect intercepts and regressors absorbed by them are identified in the table notes rather than assigned estimated zero coefficients. The predetermined factor exposures used to con-

struct NAV risk-adjusted returns are outcome inputs, not common second-stage slope regressors. The portfolio-alpha tables instead report every estimated factor loading, in dimensionless units, alongside annualized alpha.

Table A1: Samples

Analysis	Unit	Funds	Observations	Requirement
Study population	Fund	1,663		Funds with manager records, an assigned benchmark and country weights
Net returns	Fund-month	1,418	206,847	Net returns, positive lagged assets, known education, baseline 24-eligible-month cohort
Own predetermined loadings	Fund-month		174,263	At least 36 of the preceding 60 monthly returns; otherwise same-benchmark peer loadings
Broader net-return support	Fund-month	1,536	208,425	Same sources and education, including shorter observed histories with peer loadings
Joint country selection	Fund-country-quarter	1,428	974,924	All held foreign countries; education, three-month returns and team traits
Outside-country relatedness	Fund-country-quarter	1,423	936,076	No current manager schooling link; same characteristic-adjusted return
Appointments	Appointment		6,303	Recorded start, observed prior team
Appointees' prior selection	Appointment	300	1,321	Four prior payoff quarters and common benchmark-country peer support
Appointees' prior flows	Appointment	300	1,322	Prior flow and return histories
Appointee country records	Appointment-country	257	9,446	Prior country records in held destinations lacking recorded incumbent education coverage
Departures and new assignments	Manager-quarter	1,334	154,677	Known education, prior fund return
Conditional monthly flows	Fund-month	1,494	185,748	Prior returns, fund characteristics and known team education
Quarterly flows and fees	Fund-quarter	1,418	67,343	Net-return panel, next-quarter flow observed
Flows after appointments	Appointment quarter	897	2,973	Flows observed in the four quarters before and after
Country allocation	Fund-country-month	1,663	5,613,030	Fund or benchmark country weight; 1999–2024
Conviction portfolios	Portfolio-month		300	January 2000–December 2024

Notes: Every analysis starts from the 1,663 funds and then imposes its own data requirements. The joint-selection row reports 1,428 economic funds from 1,429 identifiers; the outside-country relatedness row reports 1,423 funds from 1,424 identifiers. The main NAV sample has 1,418 of each. Linked identifiers are combined for economic-fund clustering; country-allocation exhibits retain identifier clustering. Of the 1,663 funds, 1,638 appear in the return files, 1,548 have returns, 1,536 have complete quarters, positive prior assets and known education, and 1,418 meet the baseline eligible-return-history requirement. For the outside-country diagnostics, 1,623 identifiers hold foreign stocks and 1,456 have known team education before outcome and support requirements; the fit retains 1,424. Joint selection includes supported education-country observations and retains 1,429 identifiers. Missing outcomes are excluded rather than set to zero.

Table A2: Fund assets by assigned benchmark

Morningstar Index	Mean TNA (B)	Ave. Num Funds
Morningstar Gbl NR USD	4.672	51.55
Morningstar Gbl xUS Val TME NR USD	2.853	86.63
Morningstar Gbl Growth TME NR USD	2.790	51.36
Morningstar Gbl xUS Growth TME NR USD	1.787	86.45
Morningstar Gbl xUS NR USD	1.260	164.27
Morningstar EM Americas NR USD	1.091	2.57
Morningstar DM APAC xJpn NR USD	0.800	14.87
Morningstar EM NR USD	0.757	125.59
Morningstar Gbl Val TME NR USD	0.606	38.64
Morningstar Gbl xUS SMID NR USD	0.550	56.65
Morningstar DM Eur NR USD	0.528	10.16
Morningstar Gbl SMID NR USD	0.460	18.04
Morningstar Jpn NR USD	0.367	6.99
Morningstar DM APAC NR USD	0.257	3.15
Morningstar China NR USD	0.180	13.05
Morningstar India NR USD	0.121	2.34
Morningstar Gbl Lrg NR USD	0.093	28.97

Notes: This table reports the mean fund-level total net assets and the mean number of funds in each Morningstar benchmark category, computed at quarterly frequency over the 1,663-fund sample. Because the counts are per-quarter averages, the "Ave. Num Funds" column reflects funds active in a given quarter and sums to about 760—the average number of funds active per quarter—rather than the 1,663 distinct funds observed over the full sample, since funds enter and exit over time.

Table A3: Portfolio holdings

	Time-series Average			
	Mean	Median	Std	N
Sample Funds				
# of countries held	18.99	19	8.44	67,380
Country-level portfolio share(%)	4.83	2.17	9.47	1,279,345
# of stocks held	210.26	81	589.15	71,742
Stock-level portfolio share(%)	0.45	0.09	0.88	15,084,718
Morningstar Gbl xUS NR USD				
# of countries held	49.47	51	3.46	88
Country-level portfolio share(%)	1.97	0.58	3.35	4,353
# of stocks held	5,136.20	5,104	982.78	88
Stock-level portfolio share(%)	0.02	0.00	0.06	451,986
All Benchmarks (equal weights)				
# of countries held	27.55	26	20.28	1,947
Country-level portfolio share(%)	3.43	0.57	10.15	53,639
# of stocks held	1,887.58	1,344	1,841.74	1,947
Stock-level portfolio share(%)	0.05	0.01	0.23	3,675,116

Notes: This table reports country-level and stock-level holding counts and portfolio shares for the sample funds, the Morningstar Global ex-US benchmark, and the average across all Morningstar benchmarks assigned to sample funds. The sample-fund rows are computed on the same 1,663-fund sample and 2000–2024 window as Table 1. The "# held" rows are computed at the fund-quarter (or benchmark-quarter) level; the "portfolio share" rows are averaged across individual held positions (fund-country or fund-stock), so that the mean number held times the mean share equals the average total portfolio weight (≈100%, below 100 because funds also hold some U.S. equity and cash). Positions with zero weight are excluded.

Table A4: Education countries of managers

Country	# Mgrs	Share	Country	# Mgrs	Share
USA	2188	0.7313	ESP	6	0.0020
GBR	580	0.1939	NZL	6	0.0020
CAN	77	0.0257	AUT	5	0.0017
IND	70	0.0234	CHL	5	0.0017
CHN	63	0.0211	POL	4	0.0013
FRA	51	0.0170	PAK	4	0.0013
AUS	44	0.0147	VEN	4	0.0013
DEU	35	0.0117	TUR	3	0.0010
JPN	32	0.0107	ARG	3	0.0010
IRL	26	0.0087	NGA	3	0.0010
HKG	18	0.0060	NOR	3	0.0010
ZAF	16	0.0053	EGY	2	0.0007
ITA	14	0.0047	CZE	2	0.0007
SGP	12	0.0040	GRC	2	0.0007
KOR	11	0.0037	PRT	2	0.0007
CHE	11	0.0037	BGR	1	0.0003
NLD	10	0.0033	ROU	1	0.0003
DNK	10	0.0033	GTM	1	0.0003
SWE	9	0.0030	UKR	1	0.0003
BRA	8	0.0027	THA	1	0.0003
TWN	7	0.0023	EST	1	0.0003
RUS	7	0.0023	LBN	1	0.0003
MEX	7	0.0023	COL	1	0.0003
BEL	7	0.0023	IDN	1	0.0003
ISR	7	0.0023			
Total # of Managers = 2,992					

Notes: The final share column in each panel divides managers educated in the listed country by the 2,992 managers with recorded education countries (49 countries after hand review). The shares sum to more than one because a manager can have education ties to multiple countries.

Table A5: Education-country links across funds

Country	# Funds	Share	Country	# Funds	Share
USA	1492	0.898	AUT	15	0.009
GBR	639	0.384	MEX	14	0.008
IND	188	0.113	POL	14	0.008
CAN	173	0.104	BRA	12	0.007
CHN	133	0.080	VEN	8	0.005
FRA	129	0.078	NGA	8	0.005
AUS	88	0.053	CZE	8	0.005
JPN	67	0.040	PRT	8	0.005
ITA	51	0.031	CHL	7	0.004
IRL	51	0.031	EGY	7	0.004
HKG	46	0.028	ARG	6	0.004
DEU	44	0.026	GRC	6	0.004
SGP	41	0.025	PAK	6	0.004
ZAF	39	0.023	TUR	6	0.004
SWE	38	0.023	ROU	6	0.004
KOR	32	0.019	NOR	6	0.004
ISR	32	0.019	IDN	3	0.002
NLD	32	0.019	THA	2	0.001
CHE	29	0.017	LBN	2	0.001
NZL	28	0.017	GTM	2	0.001
DNK	28	0.017	UKR	2	0.001
ESP	23	0.014	EST	1	0.001
RUS	20	0.012	COL	1	0.001
TWN	19	0.011	BGR	1	0.001
BEL	16	0.010			
Total # of Funds = 1,662					

Notes: The last column reports the share of funds with at least one manager educated in each country, computed as the number of such funds divided by the total number of funds with an educated manager (1,662). This descriptive denominator includes funds with a recorded education link and does not restrict any regression sample. The shares sum to more than one because a fund can have education ties to multiple countries through its managers.

Table A6: The increasing foreign-educated appointment share

Dependent variable	Foreign appointment share Percentage points (1)
Calendar years from 2012	0.60***
	(0.18)
Log pre-hire destination team size	0.78
	(1.43)
Unknown education among appointees	−0.78
	(5.08)
Female share (known gender)	4.99
	(4.03)
Unknown-gender share	−5.10
	(6.76)
CFA share (observed biographies)	1.06
	(3.16)
Unknown-biography share	5.37
	(8.01)
Mean log(1 + experience)	−0.69
	(1.75)
Fund FE	Yes
Appointment-year FE	No
Observations	4,072
Fund clusters	1,218

Notes: The dependent variable is the foreign-educated percentage of known-education appointments. Time is measured in single calendar years from 2012. Controls describe the hiring fund's team before the appointments and unknown education among new appointees. Appointment shares are weighted by the number of known-education appointments. Errors are clustered by economic fund. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A7: Pre-appointment country relevance and net fund returns

Dependent variable	Factor-adjusted net NAV return Annualized bps	
Specification	(1) Original controls	(2) Country composition
Pre-appointment country relevance	−463.18	−688.30***
	(295.12)	(244.59)
Foreign-educated share	32.55	11.82
	(48.28)	(63.37)
Foreign share with unknown relevance	−147.74**	−224.94***
	(58.44)	(61.37)
Log team size	−87.43	−78.48
	(59.12)	(59.75)
Unknown-education share	−106.00	−84.24
	(68.38)	(68.57)
Log lagged assets (USD millions)	−21.75**	−22.49**
	(9.83)	(9.80)
Assets spline basis 1	−101.28***	−105.86***
	(25.89)	(25.73)
Assets spline basis 2	571.88***	602.09***
	(157.85)	(155.38)
Team-size spline basis 1	−80.16	−86.49
	(137.49)	(139.47)
Team-size spline basis 2	688.02	708.37
	(591.04)	(601.64)
Female share (known gender)	−56.00	−58.83
	(38.09)	(38.36)
Unknown-gender share	−72.03	−208.71**
	(91.37)	(96.61)
CFA share (observed biographies)	6.87	4.11
	(31.42)	(31.69)
Unknown-biography share	209.45*	350.28***
	(107.00)	(108.49)
Mean log(1 + experience)	−29.37*	−32.39*
	(16.59)	(16.89)
Prior-year relative NAV return (pp)	−4.63***	−4.83***
	(1.45)	(1.45)
Log(1 + prior relative volatility)	62.83**	60.31**
	(30.99)	(30.65)
Prior-quarter flow (pp)	1.55×10^{-6}	1.57×10^{-6}
	(2.25×10^{-6})	(2.27×10^{-6})
Missing prior return history	99.70	93.63
	(62.75)	(62.37)
Missing prior-quarter flow	−41.63	−41.04
	(55.61)	(55.58)
Lagged expense ratio (bps)	−0.41	−0.45
	(0.33)	(0.33)
Lagged log turnover	85.11**	88.08**
	(35.28)	(34.70)
Top-ten equity concentration	65.36	125.57
	(137.46)	(136.30)
Missing lagged expense ratio	−111.35**	−117.46**
	(53.86)	(53.49)
Missing lagged turnover	73.17**	74.54**
	(36.67)	(36.03)
Missing equity concentration	23.62	36.41
	(42.02)	(41.86)
Education-country composition controls	No	Yes
Fund FE	Yes	Yes
Benchmark × month FE	Yes	Yes
Observations	206,847	206,847
Fund clusters	1,418	1,418

Notes: Both columns augment Table 2, column 1, with pre-appointment relevance and the foreign share with unknown relevance. All degree levels are included. Appendix B.1 defines both regressors and the education-country composition controls. Relevance enters in raw units; its coefficient holds foreign representation constant. The outcome, sample, original controls and equal fund-month weights are unchanged. Standard errors clustered by economic fund appear below coefficients. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A8: Precision-weighted net fund returns

Dependent variable	Factor-adjusted net NAV return Annualized bps
Specification	(1) Precision-weighted
Foreign-educated share	-69.66^{***}
	(22.66)
Log team size	-72.93^{*}
	(37.80)
Unknown-education share	-134.06^{***}
	(44.16)
Log lagged assets (USD millions)	-17.30^{**}
	(7.38)
Assets spline basis 1	-62.79^{***}
	(18.26)
Assets spline basis 2	327.21^{***}
	(106.46)
Team-size spline basis 1	−24.61
	(88.00)
Team-size spline basis 2	394.77
	(374.04)
Female share (known gender)	−37.88
	(27.60)
Unknown-gender share	−0.34
	(53.01)
CFA share (observed biographies)	33.02
	(21.24)
Unknown-biography share	80.74
	(66.83)
Mean log(1 + experience)	-39.13^{***}
	(10.63)
Prior-year relative NAV return (pp)	-2.92^{***}
	(0.95)
Log(1 + prior rel. volatility)	36.75^{**}
	(18.59)
Prior-quarter flow (pp)	1.21×10^{-6}
	(1.51×10^{-6})
Missing prior return history	2.29
	(38.49)
Missing prior-quarter flow	-66.19^{*}
	(36.44)
Lagged expense ratio (bps)	-0.46^{**}
	(0.23)
Lagged log turnover	30.18
	(25.07)
Top-ten equity concentration	79.59
	(90.84)
Missing lagged expense ratio	−23.98
	(34.88)
Missing lagged turnover	10.76
	(23.92)
Missing equity concentration	18.74
	(27.61)
Fund FE	Yes
Benchmark × month FE	Yes
Observations	206,847
Fund clusters	1,418

Notes: The outcome is thirteen-factor-adjusted net NAV. The model retains the 206,847 observations, controls, fund effects and benchmark-month effects of Table 2, column 1, and uses inverse fund residual-variance weights. The equal-weight baseline is reported in that main table. Prior flow is measured in raw percentage points. Standard errors, clustered by economic fund, are below coefficients in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A9: Portfolio controls and return tails

Dependent variable	Factor-adjusted net NAV return Annualized bps		
Specification	(1) Style controls	(2) Portfolio controls	(3) Winsorized
Foreign-educated share	-65.55^{*}	-68.72^{*}	-68.66^{**}
	(36.51)	(35.44)	(33.08)
Log team size	−89.65	−87.40	-89.27^{*}
	(59.49)	(59.07)	(54.09)
Unknown-education share	−110.43	−106.10	−97.95
	(71.04)	(68.58)	(67.21)
Log lagged assets (USD millions)	-20.96^{**}	-21.48^{**}	-21.04^{**}
	(9.87)	(9.87)	(9.39)
Assets spline basis 1	-100.14^{***}	-97.97^{***}	-92.84^{***}
	(25.91)	(26.13)	(24.74)
Assets spline basis 2	555.51^{***}	547.88^{***}	506.05^{***}
	(158.18)	(158.47)	(149.87)
Team-size spline basis 1	−79.80	−79.90	−68.71
	(138.05)	(137.39)	(129.38)
Team-size spline basis 2	694.90	698.49	647.42
	(592.19)	(589.83)	(560.66)
Female share (known gender)	−49.50	−50.55	−43.03
	(38.37)	(38.53)	(37.14)
Unknown-gender share	−94.23	−88.81	−84.47
	(92.32)	(92.95)	(88.79)
CFA share (observed biographies)	7.61	7.26	3.59
	(31.38)	(31.45)	(30.19)
Unknown-biography share	228.55^{**}	226.85^{**}	213.12^{**}
	(107.30)	(107.68)	(102.99)
Mean log(1 + experience)	-35.24^{**}	-33.29^{**}	-34.78^{**}
	(16.95)	(16.76)	(15.21)
Prior-year relative NAV return (pp)	-4.45^{***}	-4.63^{***}	-4.20^{***}
	(1.54)	(1.47)	(1.37)
Log(1 + prior rel. volatility)	62.66^{**}	60.26^{*}	54.03^{*}
	(31.36)	(31.10)	(27.87)
Prior-quarter flow (pp)	1.57×10^{-6}	1.49×10^{-6}	1.65×10^{-6}
	(2.24×10^{-6})	(2.25×10^{-6})	(2.24×10^{-6})
Missing prior return history	73.20	71.98	79.14
	(56.93)	(59.88)	(57.92)
Missing prior-quarter flow	−44.21	−46.10	−37.84
	(55.44)	(55.55)	(53.42)
Holdings size tilt	−15.74	—	—
	(65.56)		
Holdings momentum tilt	−23.54	—	—
	(38.03)		
Missing size tilt	249.20	—	—
	(371.87)		
Missing momentum tilt	−292.73	—	—
	(346.07)		
Top-ten equity concentration	—	76.34	116.12
		(138.37)	(124.50)
Lagged log turnover	—	82.80^{**}	70.81^{**}
		(35.43)	(33.73)
Missing equity concentration	—	26.52	36.72
		(42.11)	(39.17)
Missing lagged turnover	—	40.67	57.81^{*}
		(31.75)	(33.67)
Lagged expense ratio (bps)	—	—	−0.39
			(0.32)
Missing lagged expense ratio	—	—	-95.70^{*}
			(51.41)
Fund FE	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes
Observations	206,847	206,847	206,847
Fund clusters	1,418	1,418	1,418

Notes: All columns use thirteen-factor-adjusted NAV. Columns 1–2 add style or concentration/turnover; column 3 uses full controls and 1st/99th-percentile benchmark-month winsorization. Missing indicators retain observations; raw prior flow, equal weights and fund clustering are unchanged. Table 2 reports the model without winsorization. Standard errors are below coefficients in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A10: Timing and coefficient stability of the fund-return gap

Dependent variable	Factor-adjusted net NAV return Annualized bps		
Specification	(1) Lead sample	(2) With future team	(3) Unknown share only
Foreign-educated share	-74.53^{**}	-72.28^{*}	-58.61^{*}
	(37.81)	(43.87)	(35.54)
Future foreign-educated share	—	−2.60	—
		(37.28)	
Future unknown-education fraction	—	−25.23	—
		(61.67)	
Log team size	−77.10	−77.42	—
	(60.76)	(60.79)	
Unknown-education share	-134.59^{*}	−116.77	−69.42
	(75.46)	(80.49)	(48.05)
Log lagged assets (USD millions)	-21.89^{**}	-21.86^{**}	—
	(9.95)	(9.94)	
Assets spline basis 1	-105.21^{***}	-105.21^{***}	—
	(25.92)	(25.91)	
Assets spline basis 2	587.47^{***}	587.16^{***}	—
	(157.98)	(158.01)	
Team-size spline basis 1	−111.24	−110.75	—
	(142.09)	(142.14)	
Team-size spline basis 2	824.35	822.52	—
	(612.93)	(613.13)	
Female share (known gender)	−56.32	−56.17	—
	(39.90)	(39.93)	
Unknown-gender share	−65.51	−65.84	—
	(101.26)	(101.33)	
CFA share (observed biographies)	1.45	1.44	—
	(33.24)	(33.24)	
Unknown-biography share	226.39^{*}	225.87^{*}	—
	(123.04)	(123.07)	
Mean log(1 + experience)	-39.42^{**}	-39.55^{**}	—
	(17.74)	(17.74)	
Prior-year relative NAV return (pp)	-4.78^{***}	-4.78^{***}	—
	(1.55)	(1.55)	
Log(1 + prior relative volatility)	62.86^{**}	62.82^{**}	—
	(31.62)	(31.61)	
Prior-quarter flow (pp)	1.19×10^{-6}	1.19×10^{-6}	—
	(2.27×10^{-6})	(2.27×10^{-6})	
Missing prior return history	105.34	105.20	—
	(64.36)	(64.37)	
Missing prior-quarter flow	−40.02	−40.01	—
	(56.29)	(56.30)	
Lagged expense ratio (bps)	−0.38	−0.38	—
	(0.43)	(0.43)	
Lagged log turnover	91.39^{**}	91.57^{**}	—
	(35.99)	(35.97)	
Top-ten equity concentration	48.94	49.82	—
	(143.21)	(143.24)	
Missing lagged expense ratio	-110.58^{*}	-110.62^{*}	—
	(62.49)	(62.46)	
Missing lagged turnover	60.13	60.08	—
	(37.94)	(37.93)	
Missing equity concentration	16.94	17.33	—
	(43.68)	(43.71)	
Fund FE	Yes	Yes	Yes
Benchmark × month FE	Yes	Yes	Yes
Observations	193,986	193,986	206,847
Fund clusters	1,418	1,418	1,418

Notes: Risk-adjusted net NAV is in annualized bps. Columns 1–2 require an observed team four quarters ahead; column 2 adds the future team's foreign-educated and unknown-education shares to Equation (3). Column 3 uses only those two current shares on the full sample. Against Table 2, column 1, Oster's δ for a zero coefficient is −20.6 and the bias-adjusted coefficient at Oster's $\delta = 1$ is −72.87 (Oster, 2019), with $R_{\max} = 1.3R^2$. Standard errors clustered by economic fund are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A11: Net performance beyond the initial appointment period

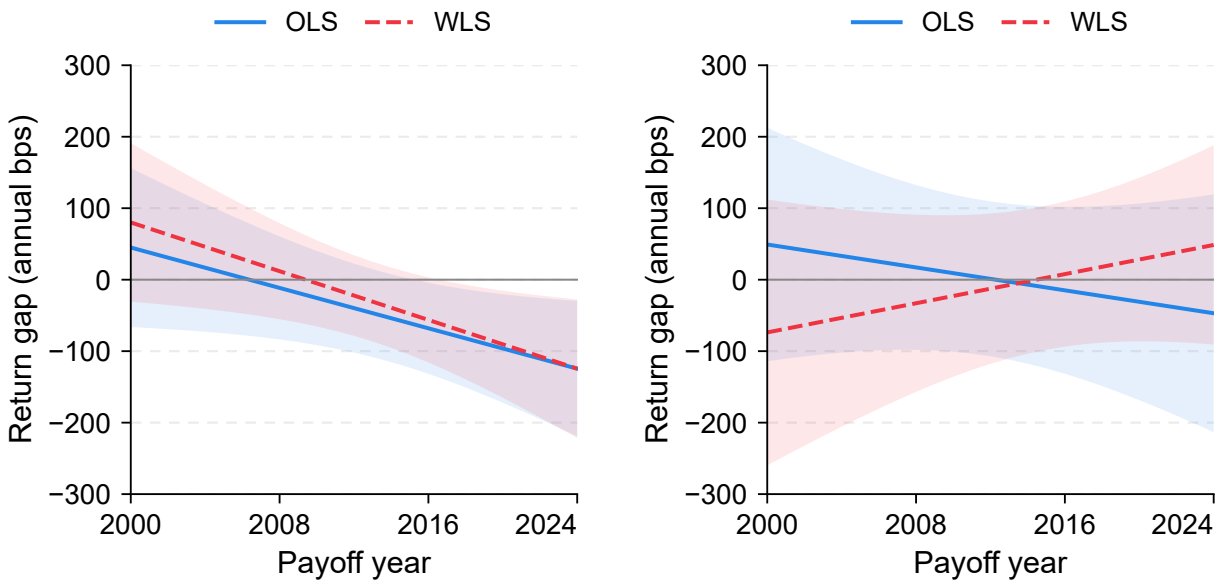
Dependent variable	Factor-adjusted net NAV return Annualized bps	
Specification	(1) Equal weights	(2) Precision weights
Foreign-educated share: established	−80.91**	−81.82***
	(38.05)	(24.28)
Foreign-educated share: first two years	−37.91	−50.15*
	(40.87)	(27.14)
U.S.-educated share: first two years	−2.56	−12.13
	(21.51)	(14.54)
Log team size	−89.98	−70.02*
	(57.87)	(37.58)
Unknown-education share	−91.04	−123.01***
	(68.25)	(43.69)
Log lagged assets (USD millions)	−16.14*	−12.65*
	(9.44)	(7.33)
Assets spline basis 1	−101.79***	−63.72***
	(26.09)	(18.25)
Assets spline basis 2	548.67***	321.81***
	(158.23)	(106.71)
Team-size spline basis 1	−71.06	−25.95
	(134.60)	(87.51)
Team-size spline basis 2	648.80	388.29
	(578.47)	(371.78)
Female share (known gender)	−51.66	−38.00
	(38.38)	(27.27)
Unknown-gender share	−95.04	−1.06
	(91.55)	(52.86)
CFA share (observed biographies)	9.43	32.91
	(31.25)	(20.91)
Unknown-biography share	230.43**	81.34
	(106.28)	(66.35)
Mean log(1 + experience)	−25.58	−33.77***
	(17.42)	(11.53)
Fund FE	Yes	Yes
Benchmark × month FE	Yes	Yes
Observations	206,847	206,847
Fund clusters	1,418	1,418

Notes: The net NAV model separates managers in their first eight quarters at the fund from established managers. Raw category coefficients use established U.S.-educated managers as reference. The first-two-years direct contrast subtracts the recent U.S.-educated coefficient from the recent foreign-educated coefficient. Outcomes use predetermined risk adjustment and reduced controls, in annualized basis points. WLS weights fund return precision. Standard errors are clustered by economic fund. OLS gives fund-months equal weight; WLS uses inverse fund residual variance. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The recent foreign-minus-U.S. contrasts are −35.35 (SE 42.14) and −38.02 (27.52), using joint covariance. Established U.S.-educated managers are the omitted category.

Table A12: Changes in the foreign-education return gap over time

Dependent variable	Adjusted country return Annualized bps	
	(1)	(2)
Specification	Equal country weights	Precision weights
Foreign-educated share × year (γ^T)	−7.06**	−8.52**
	(3.50)	(3.72)
Foreign-educated share (γ)	−39.76	−22.20
	(31.72)	(28.75)
Education-country link × year (β^T)	3.05	13.62**
	(4.98)	(5.40)
Education-country link (β)	40.89	9.61
	(48.78)	(49.80)
Log country stock count	−57.88***	−53.49***
	(11.41)	(11.19)
Log team size	−70.17***	−44.30**
	(21.50)	(18.92)
Unknown-education share	33.08	15.90
	(84.36)	(77.82)
Female share (known gender)	−90.37**	−109.49***
	(44.90)	(37.13)
Unknown-gender share	94.05	134.65
	(87.05)	(82.20)
CFA share (observed biographies)	54.13*	49.53*
	(30.10)	(28.45)
Unknown-biography share	−47.78	−68.59
	(123.27)	(111.68)
Mean log(1 + experience)	−53.72***	−47.69***
	(15.50)	(14.13)
Fund × country FE	Yes	Yes
Country × quarter FE	Yes	Yes
Observations	974,924	974,924
Fund clusters	1,428	1,428

Notes: Equation (15) adds both schooling-link and foreign-share interactions with payoff calendar year minus 2012 to the common panel in Table 6. The first coefficient is the annual change in the outside-country foreign-minus-U.S. return gap; the share coefficient gives that gap in 2012. The 2024 gaps use the entire fitted time profile and the joint coefficient covariance. Shares remain in 0–1 units; time interactions are per single year. Column 1 gives countries equal weight within each fund-quarter; column 2 additionally uses inverse fund residual variance. Both columns use fund–country and country–quarter effects, eight displayed controls and economic-fund clustering; the payoff period is 2000–2024. Standard errors are below coefficients in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Fitted 2024 outside gaps are −124.50 (SE 48.28) and −124.50 (49.38); linked-country gaps are −46.96 (84.80) and 48.59 (71.06), in columns 1 and 2. These sums use joint covariance and are plotted in Figure A1.



(a) Outside education countries

(b) Education countries

Figure A1: Education and outside-country selection gaps over time

Notes: Both panels use the joint country regressions in Table A12. Panel A plots $\gamma + \gamma^T t$; Panel B plots $\beta + \gamma + \beta^T t + \gamma^T t$, where t is years from 2012. Each compares an all-foreign-educated team with an all-U.S.-educated team; Panel B additionally requires a current manager's schooling link to the destination. Solid lines are OLS, dashed lines WLS; shaded bands are 95% intervals based on the complete economic-fund-clustered covariance matrix. The local-increment coefficient by itself is not plotted as a team-performance gap.

Table A13: Relatedness and raw country returns outside education countries

Dependent variable	Raw country return Annualized bps		
Specification	(1) Shared border	(2) Official language	(3) Social connectedness
Foreign education × tie (θ)	−345.60**	−194.62***	−81.09**
	(154.53)	(68.89)	(40.85)
Tie (δ)	188.34	132.31*	46.21
	(126.83)	(74.09)	(42.15)
Foreign-educated share (γ)	−27.33	1.41	613.09*
	(33.83)	(34.68)	(318.73)
Missing-tie share (MissingShare)	5.70	34.80	−302.82*
	(47.49)	(48.03)	(168.92)
Log country stock count	−53.07***	−53.25***	−58.83***
	(12.25)	(12.25)	(12.14)
Log team size	−85.29***	−85.81***	−80.88***
	(21.64)	(21.65)	(22.41)
Unknown-education share	−10.45	−9.43	−21.72
	(93.67)	(94.10)	(93.54)
Female share (known gender)	−76.51	−77.01	−90.77*
	(47.08)	(47.02)	(48.93)
Unknown-gender share	37.16	31.78	53.70
	(76.06)	(76.15)	(79.30)
CFA share (observed biographies)	70.73**	70.11**	74.14**
	(31.65)	(31.69)	(31.92)
Unknown-biography share	79.16	85.82	69.45
	(123.40)	(123.29)	(121.99)
Mean log(1 + experience)	−57.50***	−57.83***	−52.22***
	(16.73)	(16.76)	(17.07)
Fund × country FE	Yes	Yes	Yes
Country × quarter FE	Yes	Yes	Yes
Observations	936,076	936,076	876,532
Fund clusters	1,423	1,423	1,421

Notes: The specifications, samples, fixed effects, weights and clustering are those of Table 4; only the dependent variable differs. The raw country return is the return on the fund's holdings in the country over the three months after quarter-end, annualized in basis points, without subtracting the size and past-return matched benchmark of Equation (2). Country-quarter effects still remove the national market return common to all funds holding the country in that quarter. Column 1 is a shared border; column 2 is a common official language; column 3 is log social connectedness from the October 2021 snapshot. The first row is θ, the foreign-minus-U.S. difference in slopes; the tie coefficient δ gives the U.S.-educated slope. Standard errors in parentheses are clustered by economic fund. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A14: Foreign education and performance-conditioned monthly flows

Dependent variable	Monthly flow Percentage points (1)
Foreign-educated share	0.207
	(0.153)
Unknown-education share	−0.458
	(0.340)
Log lagged assets	−0.426***
	(0.039)
Log team size	−0.076
	(0.087)
Log fund age	−1.991***
	(0.153)
Prior-return rank: bottom segment	5.586***
	(0.605)
Prior-return rank: middle segment	1.642***
	(0.124)
Prior-return rank: top segment	8.855***
	(0.731)
Prior monthly-return volatility (pp)	−0.077*
	(0.043)
Lagged annual expense ratio (bps)	0.0004
	(0.0017)
Log(1 + turnover)	0.104
	(0.188)
Female share (known gender)	−0.361*
	(0.188)
Unknown-gender share	−0.220
	(0.381)
CFA share (observed biographies)	−0.051
	(0.138)
Unknown-biography share	0.467
	(0.494)
Mean log(1 + experience)	0.021
	(0.080)
Fund FE	Yes
Benchmark × month FE	Yes
Observations	185,748
Fund clusters	1,494

Notes: Equation (16) conditions monthly asset growth net of returns on prior performance ranks, characteristics and team traits. Prior-return rank is on a 0–1 scale, with knots at 0.2 and 0.8; the three spline terms enter the same regression. The pooled 1st/99th percentile flow caps are −21.435 and 47.589 pp. Fund age is measured from inception. Fund-months have equal weight and standard errors are clustered by economic fund. Standard errors are below coefficients in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A15: Flows and fees

Dependent variable	Next-quarter flow	Next-quarter flow	Four-quarter flow	Flow after hire	Expense ratio
	Flows in pp; expense ratio in annual bps				
	(1)	(2)	(3)	(4)	(5)
Specification	Next quarter	Return sensitivity	Four quarters	After appointment	Expense ratio
Foreign-educated share	−0.29	0.38	−0.26	—	−0.39
	(0.81)	(0.64)	(2.76)		(1.49)
Foreign-educated share × prior-year relative NAV return (pp)	—	−0.04	—	—	—
		(0.06)			
Foreign-educated share of appointees	—	—	—	0.22	—
				(1.97)	
Unknown-education share	−4.36**	−3.14**	−14.41**	—	−3.27
	(1.85)	(1.51)	(6.61)		(3.43)
Log team size	−0.40	−0.26	−2.11	—	−3.73
	(1.40)	(1.18)	(4.65)		(2.35)
Log lagged assets (USD millions)	−8.08***	−4.64***	−22.84***	−21.25***	−0.14
	(0.53)	(0.48)	(1.42)	(2.37)	(0.43)
Assets spline basis 1	3.93***	0.80	0.61	—	−5.04***
	(0.97)	(0.85)	(2.79)		(1.19)
Assets spline basis 2	−8.98*	2.09	28.74*	—	22.27***
	(5.03)	(4.27)	(15.14)		(6.92)
Team-size spline basis 1	−1.86	−1.21	−8.79	—	1.86
	(3.20)	(2.68)	(10.48)		(5.48)
Team-size spline basis 2	10.81	7.75	55.47	—	9.70
	(13.43)	(11.30)	(43.82)		(22.90)
Female share (known gender)	−0.72	−1.38*	−5.29	1.65	−1.14
	(1.07)	(0.83)	(3.55)	(4.28)	(1.61)
Unknown-gender share	−1.11	−1.04	−2.38	−6.94	−1.41
	(1.93)	(1.46)	(6.77)	(12.95)	(2.52)
CFA share (observed biographies)	−0.88	−0.71	−3.65	−3.33	−1.33
	(0.73)	(0.60)	(2.44)	(4.38)	(1.26)
Unknown-biography share	3.12	1.97	8.18	15.67	4.87
	(2.51)	(2.08)	(9.20)	(18.28)	(4.02)
Mean log(1 + experience)	−1.59***	−0.76**	−2.70**	0.24	−1.24**
	(0.40)	(0.33)	(1.27)	(2.41)	(0.61)
Prior-year relative NAV return (pp)	—	0.38***	—	0.68***	—
		(0.03)		(0.17)	
Pre-appointment four-quarter flow (pp)	—	—	—	0.07*	—
				(0.04)	
Log pre-hire destination team size	—	—	—	0.55	—
				(2.62)	
Unknown education among appointees	—	—	—	−1.05	—
				(4.80)	
Fund FE	Yes	Yes	Yes	Yes	Yes
Benchmark × quarter FE	Yes	Yes	Yes	No	Yes
Year FE	No	No	No	Yes	No
Observations	67,343	63,869	62,657	2,973	61,639
Fund clusters	1,418	1,418	1,418	897	1,418

Notes: Flows are net flows as percentages of assets; the final column uses annual expense-ratio basis points. Columns 1–2 explain next-quarter flows, column 3 the next four quarters, and column 4 the four quarters following an appointment conditional on prior flows. Column 2 adds the interaction of foreign-educated share with prior relative performance, in annual NAV percentage points. Team information precedes subsequent flows. All five regressions use equal observation weights and economic-fund clustering. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A16: Prior records relative to benchmark-country-quarter peers

Dependent variable	Peer-adjusted prior return Annualized basket bps (1)
Foreign-educated appointee	−71.77*
	(40.96)
Log pre-hire destination team size	−13.85
	(66.08)
Pre-hire destination unknown education	422.18**
	(206.91)
Log origin team size	−43.59
	(55.09)
Origin-team unknown education	161.62
	(330.54)
Log origin country stock count	−68.35
	(77.72)
Appointee: top-100 school	54.35
	(34.48)
Appointee: school ranked 101–500	134.55**
	(56.31)
Appointee: female	−18.41
	(38.26)
Appointee: unknown gender	43.13
	(70.74)
Appointee: CFA	−45.16
	(35.06)
Log(1 + manager experience)	22.20
	(31.79)
Prior benchmark: China	70.61
	(305.31)
Prior benchmark: Developed Asia-Pacific	−1029.96
	(953.70)
Prior benchmark: Developed Asia-Pacific ex Japan	32.28
	(179.00)
Prior benchmark: Developed Europe	492.95**
	(234.47)
Prior benchmark: Emerging Americas	707.99**
	(338.04)
Prior benchmark: Emerging markets	31.50
	(128.52)
Prior benchmark: Global growth	191.20
	(237.58)
Prior benchmark: Global large cap	127.49
	(212.64)
Prior benchmark: Global	96.97
	(164.44)
Prior benchmark: Global small/mid cap	102.20
	(270.14)
Prior benchmark: Global value	20.29
	(220.97)
Prior benchmark: Global ex-U.S. growth	−66.31
	(118.35)
Prior benchmark: Global ex-U.S. small/mid cap	134.34
	(145.04)
Prior benchmark: Global ex-U.S. value	−23.98
	(101.81)
Prior benchmark: Japan	908.97
	(2120.85)
Hiring-fund FE	Yes
Appointment-year FE	Yes
Observations	1,321
Hiring-fund clusters	300

Notes: Equation (11), with the peer-adjusted prior team record as outcome. The same 1,321 appointments and 27 regressors as Table 9, column 1, are retained, but the outcome covers outside-country holdings only. Other economic funds sharing the origin benchmark, country and quarter supply peers. All prior benchmark shares are shown; Global ex-U.S. is the reference. No prior NAV control. Equal appointment weights; hiring-economic-fund clustering. Standard errors are below coefficients in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A17: Accepted records within the same hiring round

Dependent variable	Prior adjusted return Annualized basket bps (1)
Foreign-educated appointee	−80.52*
	(41.29)
Log origin team size	69.06
	(91.83)
Origin-team unknown education	−546.09**
	(260.01)
Log origin country stock count	−126.69
	(138.88)
Appointee: top-100 school	−17.06
	(27.92)
Appointee: school ranked 101–500	23.94
	(27.75)
Appointee: female	−17.97
	(30.38)
Appointee: unknown gender	270.17**
	(105.25)
Appointee: CFA	−14.23
	(25.84)
Log(1 + manager experience)	78.88***
	(29.10)
Hiring-fund × appointment-quarter FE	Yes
Observations	832
Hiring-fund clusters	189

Notes: The outcome is the accepted recruit's four-quarter prior origin-team selection record, annualized in basket bps. The model compares 832 appointments in 271 multi-appointee rounds within the same destination fund and appointment quarter; 126 rounds include both education groups. Destination pre-hire team size and unknown schooling are absorbed by the fixed effects. Appointments are equally weighted, with destination-economic-fund clustering. The broader record comparison is in Table 9. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%.

Table A18: Country coverage and relative departure responses

Dependent variable	Departure Probability in pp (1)
Unique coverage × fund return (pp)	0.180***
	(0.056)
Duplicate coverage × fund return (pp)	0.006
	(0.118)
No match × fund return (pp)	0.122**
	(0.062)
Unclassified coverage × fund return (pp)	0.051
	(0.157)
Unique held-country coverage	−0.12
	(0.41)
Duplicate held-country coverage	−0.02
	(0.78)
Unclassified held-country coverage	−0.83
	(0.81)
Log recorded fund spell length	2.70***
	(0.28)
Female × fund return (pp)	0.067
	(0.056)
Unknown gender × fund return (pp)	0.012
	(0.138)
CFA × fund return (pp)	−0.051
	(0.044)
Unknown biography × fund return (pp)	−0.111
	(0.249)
Log(1 + experience) × fund return (pp)	0.045
	(0.029)
Log(1 + manager experience)	−3.16***
	(0.41)
Fund × quarter FE	Yes
Fund × manager FE	Yes
Observations	154,677
Fund clusters	1,334

Notes: Coverage is measured using prior holdings and incumbent colleagues: unique, duplicated, unmatched or unclassified. Return interactions and their direct differences are percentage points per percentage point of better prior-quarter performance. Downside examples are calculated in the text. These are relative foreign-minus-U.S. exit responses. Managers receive inverse team-size weights, and standard errors are clustered by economic fund. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Duplicate minus unique and no-match minus unique return slopes are −0.175 (SE 0.119) and −0.058 (0.076). Jointly testing the duplicate and no-match slopes equal to zero gives $F = 1.96$, $p = 0.141$.

Table A19: Conviction portfolios: alternative constructions

Dependent variable	Portfolio excess return							
	Annual alpha in bps; factor loadings dimensionless							
Specification	(1) B-measure	(2) U-measure	(3) Ex-U.K.	(4) Undergrad.	(5) Momentum	(6) Mom., no bm.	(7) Benchmark only	(8) Net costs
Alpha (annualized bps)	275.4*	318.3**	574.1***	492.8***	380.5***	369.8***	379.9***	341.4***
	(147.5)	(132.1)	(148.0)	(134.3)	(136.8)	(136.8)	(124.8)	(124.8)
Global ex-U.S. benchmark excess return	0.101	0.158	0.095	0.182*	0.146	—	1.004***	0.184*
	(0.117)	(0.105)	(0.117)	(0.106)	(0.107)		(0.021)	(0.099)
Developed ex-U.S. market	0.790***	0.775***	0.781***	0.752***	0.780***	0.902***	—	0.749***
	(0.107)	(0.096)	(0.107)	(0.097)	(0.098)	(0.043)		(0.090)
Developed ex-U.S. size	−0.079	−0.065	0.020	−0.001	−0.015	−0.015	—	0.007
	(0.066)	(0.059)	(0.066)	(0.060)	(0.062)	(0.062)		(0.055)
Developed ex-U.S. value	−0.051	−0.028	−0.100	−0.092	−0.001	0.012	—	−0.015
	(0.076)	(0.068)	(0.076)	(0.069)	(0.072)	(0.071)		(0.064)
Developed ex-U.S. investment	−0.104	0.070	0.021	0.087	0.020	0.017	—	−0.073
	(0.103)	(0.092)	(0.104)	(0.094)	(0.096)	(0.097)		(0.087)
Developed ex-U.S. profitability	−0.125	0.052	−0.156	−0.092	0.011	0.025	—	−0.024
	(0.102)	(0.092)	(0.103)	(0.093)	(0.093)	(0.093)		(0.087)
Emerging-market market	0.176***	0.101**	0.162***	0.101**	0.104**	0.128***	—	0.108***
	(0.048)	(0.043)	(0.048)	(0.044)	(0.044)	(0.040)		(0.041)
Emerging-market size	0.044	0.009	−0.056	0.014	−0.004	−0.014	—	0.015
	(0.071)	(0.064)	(0.071)	(0.065)	(0.065)	(0.064)		(0.060)
Emerging-market value	−0.113	−0.029	0.002	−0.034	−0.012	−0.013	—	−0.069
	(0.073)	(0.065)	(0.073)	(0.066)	(0.066)	(0.066)		(0.062)
Emerging-market investment	0.203**	−0.027	−0.080	−0.015	−0.046	−0.048	—	0.082
	(0.085)	(0.076)	(0.086)	(0.078)	(0.078)	(0.078)		(0.072)
Emerging-market profitability	0.125	−0.081	−0.100	0.040	−0.076	−0.080	—	0.022
	(0.094)	(0.084)	(0.094)	(0.085)	(0.087)	(0.087)		(0.079)
Developed ex-U.S. momentum	—	—	—	—	0.001	0.007	—	—
					(0.043)	(0.043)		
Emerging-market momentum	—	—	—	—	0.025	0.021	—	—
					(0.050)	(0.050)		
Observations	300	300	300	300	300	300	300	300

Notes: Each column reports one top-portfolio factor regression. Annual alpha is $12 \times 10^4 \alpha_p$ basis points; loadings are dimensionless. The B-measure uses benchmark weights over held stocks, and the U-measure expands that reference set to the preceding eight quarters. The last two constructions use only the benchmark, or deduct estimated trading costs. All use January 2000–December 2024 and unchanged portfolio weights. Portfolio standard errors are conventional OLS; direct Wald tests use the robust stacked covariance. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%. Top-versus-ordinary Wald p-values for the eight columns are 0.251, 0.057, 0.007, < 0.001, 0.025, 0.025, 0.024 and 0.003, using the robust stacked covariance.

Table A20: Local alpha under common source support and alternative weights

Dependent variable	Portfolio excess return					
	Annual alpha in bps; factor loadings dimensionless					
Specification	(1) Common mix: top	(2) Common mix: ordinary	(3) Common mix: combined	(4) Actual: top	(5) Actual: ordinary	(6) Actual: combined
Alpha (annualized bps)	406.7***	3.3	130.8	16.6	−177.5	−103.2
	(134.1)	(92.2)	(94.9)	(159.3)	(109.1)	(119.2)
Global ex-U.S. benchmark excess return	0.155	0.116	0.137*	0.249**	0.117	0.180*
	(0.106)	(0.073)	(0.075)	(0.126)	(0.086)	(0.094)
Developed ex-U.S. market	0.777***	0.731***	0.749***	0.895***	0.858***	0.872***
	(0.097)	(0.067)	(0.069)	(0.115)	(0.079)	(0.086)
Developed ex-U.S. size	−0.007	−0.138***	−0.085**	−0.155**	−0.326***	−0.248***
	(0.060)	(0.041)	(0.042)	(0.071)	(0.048)	(0.053)
Developed ex-U.S. value	−0.008	0.050	0.036	0.170**	0.136**	0.158**
	(0.069)	(0.047)	(0.049)	(0.082)	(0.056)	(0.061)
Developed ex-U.S. investment	0.008	−0.004	0.009	0.018	−0.026	−0.008
	(0.094)	(0.065)	(0.066)	(0.111)	(0.076)	(0.083)
Developed ex-U.S. profitability	0.007	0.033	0.050	0.267**	0.167**	0.220***
	(0.093)	(0.064)	(0.066)	(0.111)	(0.076)	(0.083)
Emerging-market market	0.101**	0.176***	0.138***	−0.113**	0.040	−0.031
	(0.044)	(0.030)	(0.031)	(0.052)	(0.036)	(0.039)
Emerging-market size	0.000	−0.037	−0.024	0.158**	0.121**	0.139**
	(0.065)	(0.045)	(0.046)	(0.077)	(0.053)	(0.058)
Emerging-market value	−0.020	−0.020	−0.016	−0.009	0.071	0.037
	(0.066)	(0.046)	(0.047)	(0.079)	(0.054)	(0.059)
Emerging-market investment	−0.018	−0.075	−0.056	0.057	−0.039	0.008
	(0.078)	(0.053)	(0.055)	(0.092)	(0.063)	(0.069)
Emerging-market profitability	−0.028	−0.006	0.001	0.012	−0.040	−0.014
	(0.085)	(0.059)	(0.060)	(0.101)	(0.069)	(0.076)
Observations	300	300	300	300	300	300

Notes: All portfolios use the same 77,188 fund-country-month cells at 749 identifiers and 300 payoff months. Columns 1–3 use benchmark country shares and columns 4–6 actual capital. Alpha is annualized; factor loadings are dimensionless monthly slopes with conventional OLS standard errors. The combined actual-minus-country-reweighted alpha uses the joint covariance. Different top/ordinary country mixtures and time-varying capital shares prevent combining their alphas using a single average weight. Standard errors are in parentheses. *, ** and *** denote two-sided significance at 10%, 5% and 1%. The combined actual-minus-common-mix alpha, column 6 minus column 3, is −233.98 bps (SE 85.45, $p = 0.007$), using the joint covariance.

Table A21: Principal variables, units and timing

Variable	Unit	Definition and observation window
Foreign-educated manager share ForeignShare_{jq}	Fraction of known-education managers	Any non-U.S. post-secondary education, including dual U.S./foreign degrees. Quarter-end team used for the following quarter's returns. Measures education location, not nationality or diversity.
Unknown-education share UnknownShare_{jq}	Fraction of all current managers	Managers without identifiable education divided by the recorded team count; entered separately from ForeignShare.
Education-country link	Country indicator or team share	Current managers' recorded education ties to the destination. An outside destination has no such tie.
Factor-adjusted fund return, R^a_{jm}	Annualized fund NAV basis points	Monthly net NAV less the risk-free rate and thirteen predetermined factor-exposure returns. Uses preceding returns and the stated peer fallback; covers the entire fund.
Adjusted country return, R_{jcq}	Annualized basis points	Equation (2): twelve times the mean of the three monthly size/past-return-adjusted returns after quarter-end. Positions drift within the quarter; stocks without returns are dropped. Team and holdings then refresh.
Appointee's prior record, $\bar{R}^{\text{pre}}_h$	Annualized country-basket basis points	Origin-team selection over four full pre-hire payoff quarters. Equal supported-country weights, then averages over quarters and distinct economic origin funds. The main hiring comparison includes all non-U.S. countries and controls origin benchmarks; the $\bar{R}^{\text{peer}}_h$ check uses outside countries only and subtracts benchmark-country-quarter peers. A team record, not personal NAV.
Career performance signal, R^{rel}_{jq}	Quarter-q return in percentage points	Compounded NAV in q relative to other funds sharing the benchmark; predicts careers between the ends of q and $q+1$. Distinct from prior-year controls in the NAV and flow models.
Career outcomes, $\mathbb{1}^{\text{Exit}}_{ijq}$ and $\mathbb{1}^{\text{NewFund}}_{ijq}$	Probability in percentage points	Departure from q to $q+1$, or new recorded appearance at another economic fund in $q+1$. Assignments need not be promotions.
Country-coverage need	Pre-hire matched holdings	Prior-quarter education-country exposure and continuing incumbents' ties. Missing coverage lacks an incumbent tie; no match means no held education country. Missing books remain unclassified.
Conviction, I_{jkq}	Within-country active weight	Stock weight less capitalization weight in the same held-country basket. Quartiles precede future returns; portfolio alphas use monthly country-reweighted, factor-adjusted returns.

Notes: The sources, fixed effects, controls and samples of each comparison are described with the corresponding analysis. Panel regressions cluster standard errors by economic fund, except the country-allocation regressions, which cluster by fund identifier; calendar-time conviction portfolios use the time-series covariance specified in Table 7. Education comes from retrospectively collected biographies assigned to dated fund spells.